



PROVISIONING AND INSTALLATION GUIDE OF THE K2VIEW AZURE MARKETPLACE APPLICATION

K2view Data Product Platform Services

Version: 1.3

Date: 6 November 2024

This document contains copyrighted work and proprietary information belonging to K2view. All intellectual property rights (including, without limitation, copyrights, trade secrets, trademarks, etc.) evidenced by or embodied in and/or attached, connected, or related to this document, as well as any information contained herein, are and shall be owned solely by K2view. K2view does not convey to you an interest in or to this document, the information contained herein, or its intellectual property rights, but only a personal, limited, fully revocable right to use the document solely for review purposes. Unless explicitly set forth otherwise, you may not reproduce by any means any document and/or copyright contained herein.

Information in this document is subject to change without notice. Corporate and individual names and data used in the examples herein are fictitious unless otherwise noted.

Copyright © 2024 K2view Ltd. / K2VIEW LLC. All rights reserved.

The following are trademarks of K2view: K2view logo, K2view's platform.

K2view reserves the right to update this list document from time to time.

Overview

This guide provides a comprehensive walkthrough for provisioning and installing the K2view Data Product Platform in a self-hosted environment on Microsoft Azure. Targeted for IT and DevOps audiences, it details each phase of the installation process - from the steps and prerequisites to install the K2view Azure Application resources to deploying and configuring the K2view's cloud services within your self-hosted environment. The guide addresses setting up Kubernetes clusters, managing resource groups, and ensuring connectivity to customer-managed services.

This documentation outlines necessary prerequisites, roles, and post-installation validation steps. Please use the prerequisites and installation checklist to prepare and plan your installation.

Should a problem arise during the installation, please consult with your K2view representative to obtain support.

Table of Contents

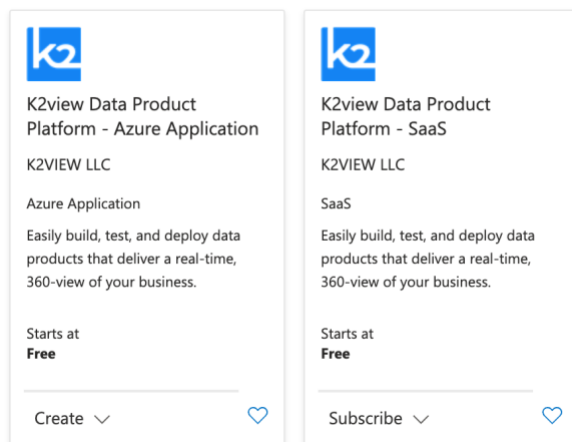
Overview.....	2
1. The End State	4
2. Planning.....	8
2.1. Steps.....	8
2.2. Constraints.....	9
2.3. Prerequisites.....	10
2.4. Installation Checklist	14
2.5. Installation Worksheet	15
2.6. Post Installation	18
3. Installing the K2view Azure Application	19
4. Populating Your Azure Container Registry.....	29
4.1. Getting Acquainted with the K2view Nexus Repository	29
4.2. Commands to Pull Containers from the K2view Nexus Repository	30
4.3. Commands to Push Containers to the Azure Container Registry.....	31
4.4. Selecting a Different Azure Container Registry or OCI-Compliant Registry	33
4.5. Sharing Information with K2view.....	33
5. Creating Your Fabric Project	34
5.1. Sign In to K2cloud.....	34
5.2. Create the Project	35
6. Creating Your First Fabric Space	38
7. Validation	40
7.1. Enter The Fabric Space	40
8. Completion.....	41
9. Instructions	42
9.1. Azure Resource Registration	42
9.2. GitHub	46
9.3. DNS zone	51
9.4. Certificates.....	53
10. Troubleshooting	57

1. The End State

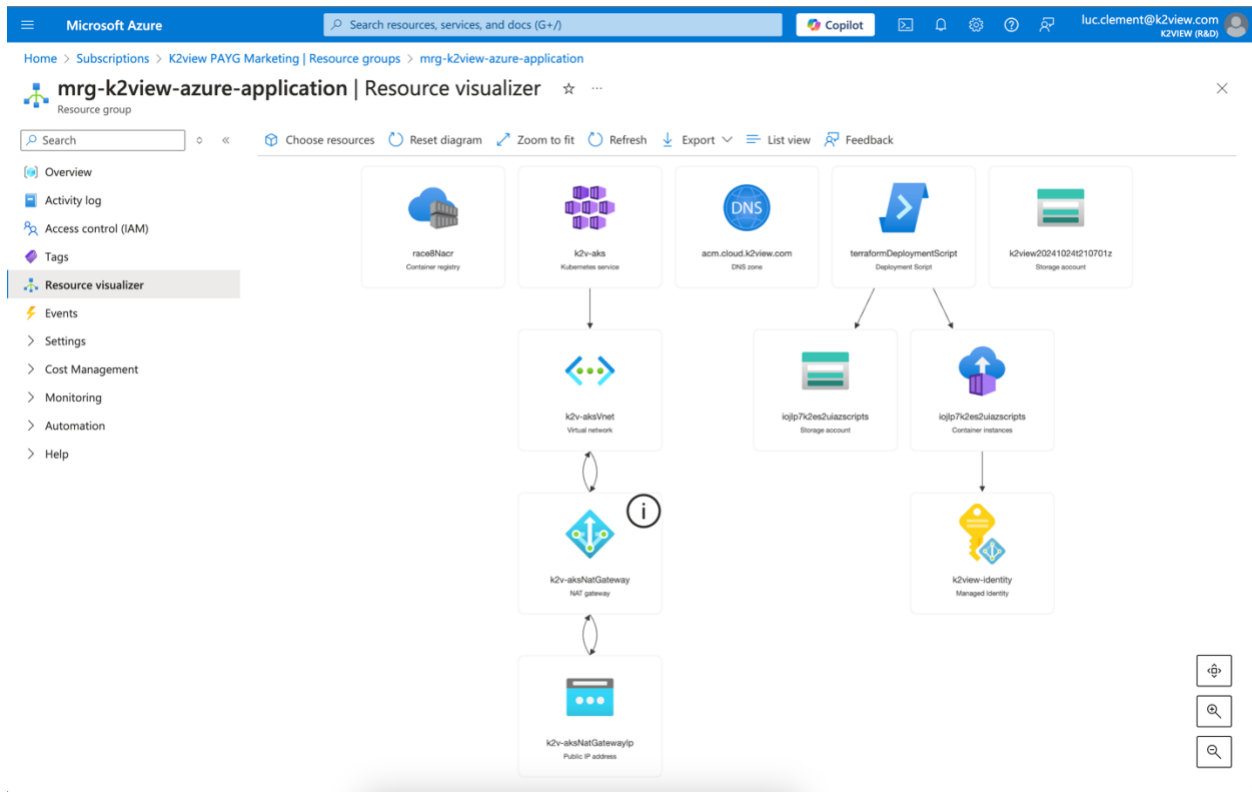
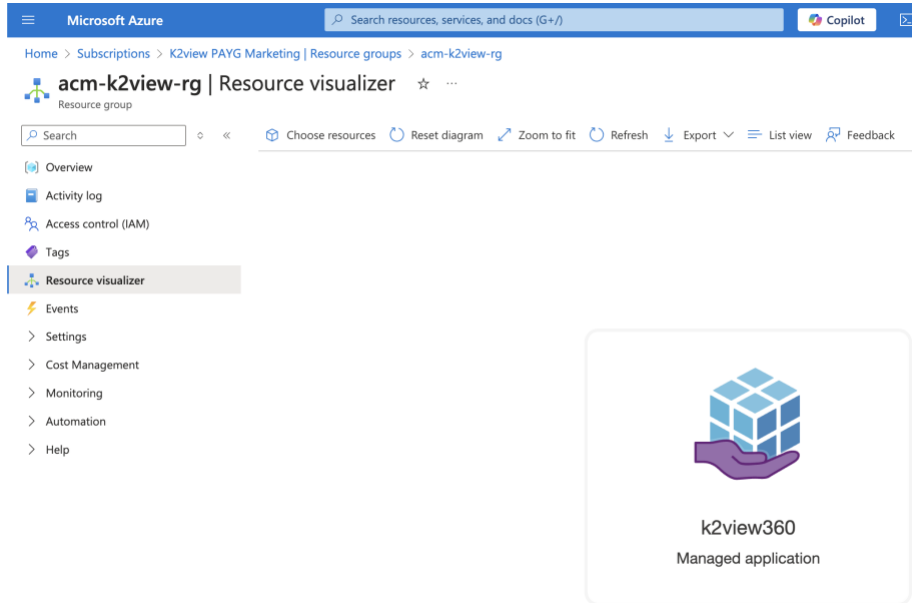
Upon completing the instructions in this guide, the K2view Data Product Platform will be fully deployed and operational within your Azure environment. This includes a fully configured Azure Kubernetes Service (AKS) cluster, secured network components, and connected resource groups, allowing you to start building and managing data products. The installed environment will be ready for integration with your existing data sources and K2view's cloud services, enabling the creation of projects and spaces for data management. Additionally, your Azure Container Registry will be populated with the necessary K2view Docker images, setting the foundation for scalable and secure operations

The [Planning](#) section will guide you with the information you need to gather to complete the installation.

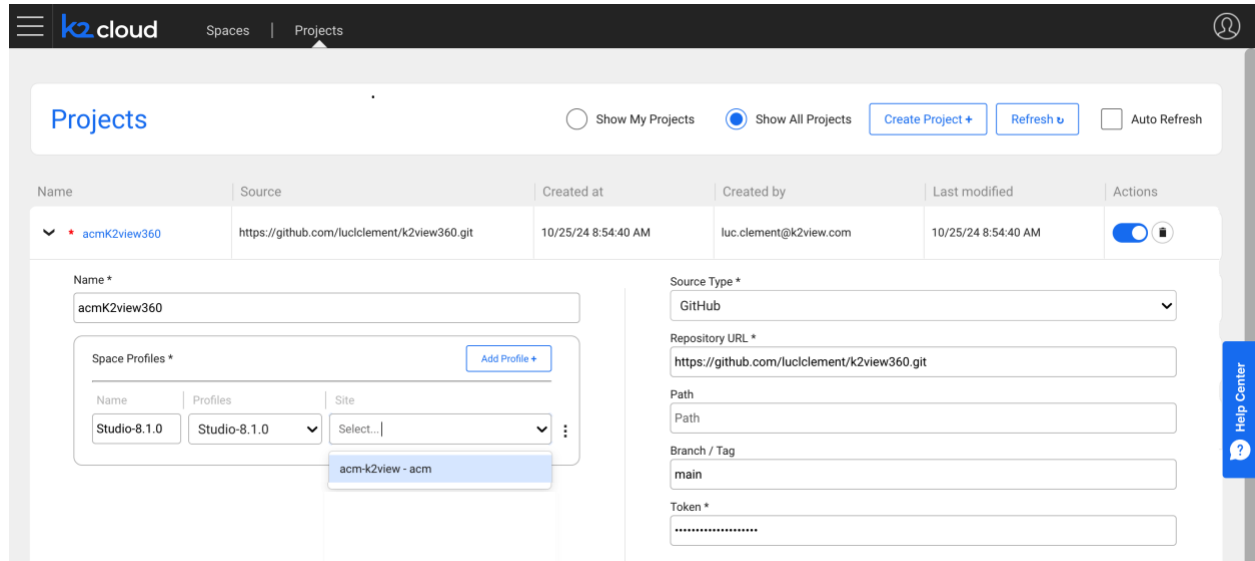
The process starts by selecting the **K2view Data Product Platform—Azure Application** listing on the Microsoft Azure Marketplace. This listing will be used to install the K2view Data Product Platform within your Azure environment.



Upon completing the installation two resource groups will have been created: one for the K2view Managed Application and the other for the Kubernetes and infrastructure services on which the K2view Data Product Platform services will run.



Upon completion of the installation, you will create your K2view project.



The screenshot shows the 'Projects' page in the K2view cloud interface. The project 'acmk2view360' is selected. The details form shows the following information:

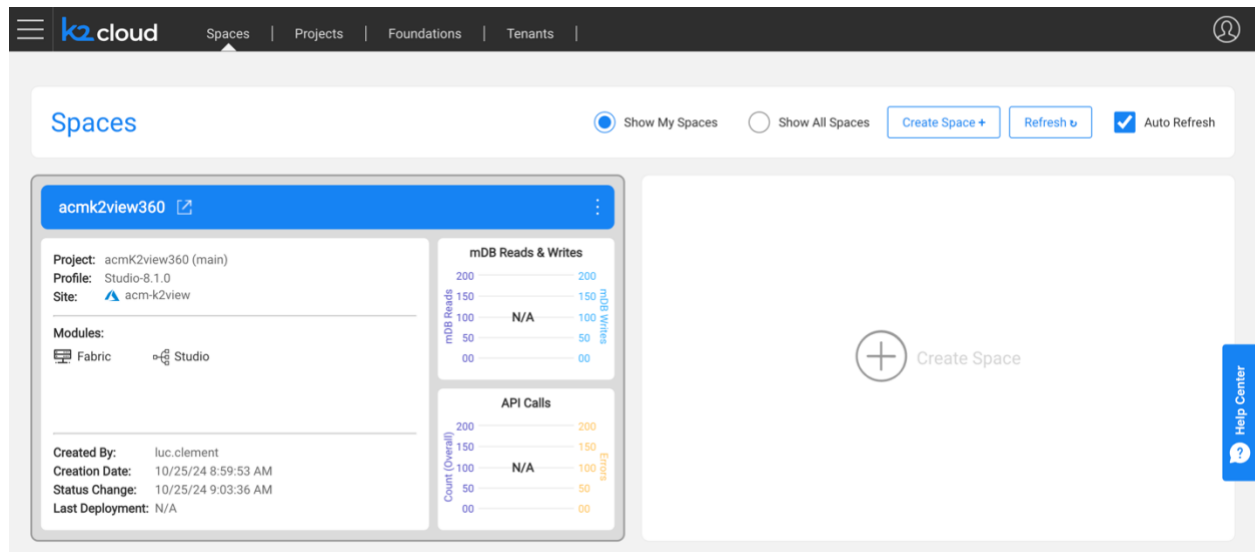
- Name:** acmk2view360
- Source:** https://github.com/luccllement/k2view360.git
- Created at:** 10/25/24 8:54:40 AM
- Created by:** luc.clement@k2view.com
- Last modified:** 10/25/24 8:54:40 AM
- Source Type:** GitHub
- Repository URL:** https://github.com/luccllement/k2view360.git
- Path:** Path
- Branch / Tag:** main
- Token:** *****

The 'Space Profiles' section shows a table with the following data:

Name	Profiles	Site
Studio-8.1.0	Studio-8.1.0	Select...

A dropdown menu is open for the 'Site' column, showing the option 'acm-k2view - acm'.

And then, you will create your first K2view Space.



The screenshot shows the 'Spaces' page in the K2view cloud interface. The space 'acmk2view360' is selected. The details form shows the following information:

- Project:** acmk2view360 (main)
- Profile:** Studio-8.1.0
- Site:** acm-k2view
- Modules:** Fabric, Studio
- Created By:** luc.clement
- Creation Date:** 10/25/24 8:59:53 AM
- Status Change:** 10/25/24 9:03:36 AM
- Last Deployment:** N/A

The 'mDB Reads & Writes' section shows a bar chart with the following data:

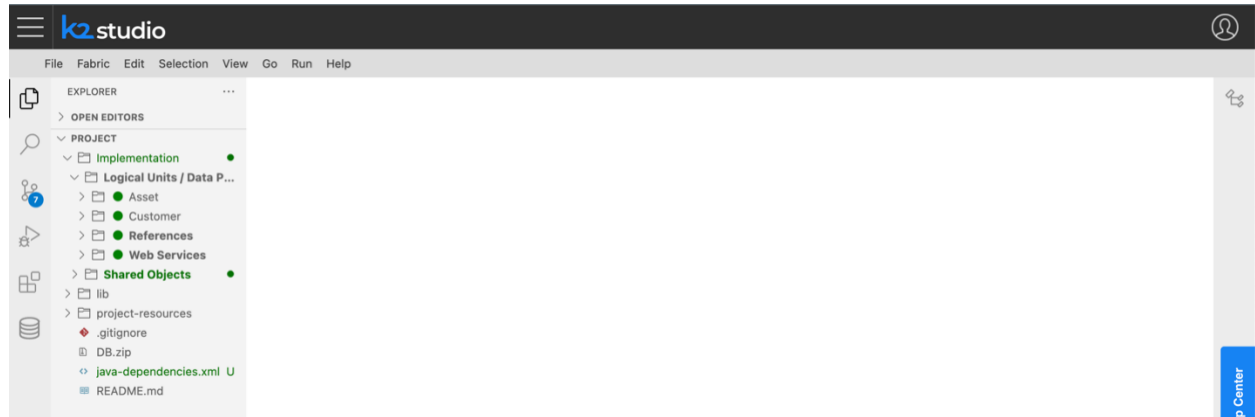
mDB Reads	mDB Writes
N/A	N/A

The 'API Calls' section shows a bar chart with the following data:

Count (Overall)	Errors
N/A	N/A

A large 'Create Space' button is visible on the right side of the page.

And finally, you will be ready to start your first project, and start learning using our Knowledge Base at <https://support.k2view.com/knowledge-base.html>, where you can find documentation and demo projects, and sign up for the K2view Academy Learning Center at <https://academy.k2view.com>.



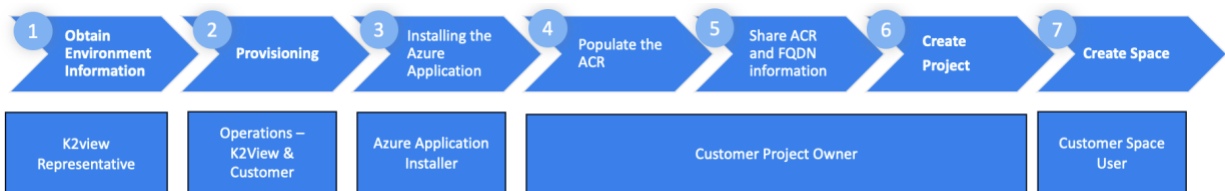
Next, let's plan this installation.

2. Planning

Planning and capturing the necessary information prior to the installation will help ensure a smooth installation experience. The customer and K2view will gather and share information used to provision the environment.

This section summarizes the steps, constraints, and prerequisites. At the end of the section, you will find an [Installation Checklist](#) and an [Installation Worksheet](#) that will help you ensure you have everything on hand to complete the installation.

2.1. Steps



Step 1 – Obtaining Environment Information

A K2view representative working with the customer will discuss the prerequisites to complete the installation and review the [Installation Checklist](#).

Step 2 - Provisioning

During the provisioning step:

- K2view will create Nexus and user accounts, create a tenant, and create associated configurations. Some of the information that K2view needs to complete the configuration will need to be provided by the customer in Step 5.
- The customer will gather information based found in the [Installation Checklist](#) such as Git details, a TLS certificate (PEM encoded) and its private key (PEM encoded), and names of entities. The certificate and private key files need to be base64 encoded. The customer will also need to validate prerequisites for the user installing the K2view Azure Application within the Azure subscription.

Step 3 – Install the K2view Azure Application

A person with the necessary roles – the Subscription Owner role - within the Azure subscription will run the [K2view Azure Application installation](#).

Step 4 – Populate the Azure Container Registry with K2view Docker images

The K2view project owner will [load the selected Azure Container Registry](#) with K2view Docker images that have been identified as required for the customer during Step 1.

Step 5 – Share ACR and Domain Name information with K2view

To complete the configuration for the tenant's creation of Projects and Spaces, the Customer Project Owner will need to share the following information used.

- The fully qualified domain name (FQDN) used for your Domain Ingress
- The list of Azure Container Registry (ACR) locations of your Docker images
- The AKS Cluster name

This information may have been identified during pre-planning, but the specific information should be confirmed to avoid misconfiguration.

Step 6 - [Create the K2view Project](#)

Step 7 - [Create a K2view Space](#)

2.2. Constraints

There are three constraints you need to be aware of to install the K2view Azure Application within your Azure subscription.

- **Azure Subscription Type.** You will be installing the K2view Azure Application in an Azure Subscription that needs to be a Pay-As-You-Go (PAYG) or Enterprise Agreement subscription.
- **Subscription Owner Role.** The installer of the K2view Azure Application needs specific Subscription level roles. The user needs to be a **Subscription Owner**.
- **Azure Resource Registration.** Azure resources must be registered before the installation by a **Subscription Owner**. The list of resources and commands to use are provided in the [Azure Resource Registration](#) section.
- **Region and availability zones.** Please note that the installation requires you select an Azure region that offers more than one availability zones. Not all Azure Regions support multiple availability zone.

2.3. Prerequisites

Certain prerequisites must be met, and planning steps must be taken to install the K2view Azure Application, set up the Azure Container Registry, and create a K2view Project and a Space.

These prerequisites are summarized here.

- You need to verify that your Azure Subscription is a PAYG or an Enterprise Agreement subscription.
- You need the person performing the K2view Azure Application installation to have a **Subscription-level Owner** role.
- Azure resources must be registered before the installation by a **Subscription Owner** (see: [Azure Resource Registration](#)).
- A Git code repository is required for your Fabric Project. Three pieces of information will be required to create your K2view project including:
 - Repository URL
 - Git token
 - Git Branch / Tag

The creation of a Git repository, if one is not already available for your use, is described in the [GitHub](#) section.

- You will need access to K2view's Nexus Repository and Docker installed on a machine used to populate the Azure Container Registry (ACR) that will be created within your Azure Resource Group.
 - The installation of the K2view Azure Application will create an ACR; if you plan to use a pre-existing ACR you may do so. Whether you use the ACR or an OCI-compatible container registry the location where you will be populating K2view Docker images needs to be provided to K2view as described in Step 5 above and in the [Sharing information with K2view](#) section.
 - You need to populate your selected ACR by pulling and pushing docker images.
 1. Docker needs to be installed on a computer to pull and push docker images from the K2view Nexus repository to the ACR created
 2. You will need a K2view Nexus account you can obtain from your K2view representative

The [Populating Your Azure Container Registry](#) section describes the steps.

- Pre-planning names will help a smooth installation. Use the [Checklist](#) and capture this information in a document. This includes:
 - The Azure region to be used. Please note that the installation requires an Azure Region that provides two or more availability zones. Not all Azure Regions have multiple availability zones.
 - The names of the two resource groups where K2view Fabric services will be installed.
 1. **Azure Application Resource Group**. You can choose an existing resource group or create a new one. You should select a representative name. In the example

demonstrated in this guide we use “acm-k2view-rg”. Select something that has significance to your project.

2. **Azure Managed Application Resource Group.** K2view resources will be created in a separate resource group, the Azure Managed Application resource group. This resource group contains K2view infrastructure components created during the installation. A name will be suggested for you. You can use that name or one that has significance to you. In our example, we used **mrg-k2view-azure-application**.

- **Cluster Name:** This is an Azure Kubernetes Service (AKS) cluster name of the Kubernetes cluster to be created. The default, if used, is k2v-aks.
- **Environment Tag:** This tag describes the project's environment. Typical values could be Dev, UAT, and Prod. Azure cost management modules will use this tag to help you categorize costs.
- **Project Tag:** This tag describes the functional purpose of the project. You might consider tag names such as K2view TDM, K2view Masking, and K2view 360. Azure cost management modules will use this tag to help you categorize costs.
- **Domain Ingress:** This is the FQDN of the Nginx Ingress Controller for which a PEM wildcard certificate needs to be created. In our example, we used acm.cloud.k2view.com, a domain owned by K2view. You will need to select your own domain and one for which you can create TLS certificates (PEM encoded).

- TLS keys and certificates

Your network team must create a wildcard TLS certificate for the domain you selected as your Domain Ingress. The full certificate authority chain needs to be present. During the installation, you will need to paste the base64 values of your PEM TLS certificate and its associated private key.

- **TLS Private Key:** the base64-encoded PEM private key associated with the TLS certificate
- **TLS Certificate:** a base64-encoded PEM wildcard certificate. For example, for the domain ingress of acm.cloud.k2view.com you need to create a wildcard certificate for *.acm.cloud.k2view.com for hosts of this domain.

These should look like the following, respectively (shortened for brevity and non-disclosure of sensitive):

PEM TLS Private Key:

```
-----BEGIN PRIVATE KEY-----
MIIJQwIBADANBgkqhkiG9w0BAQEFAASCSS0wggkpbAgEAAoICAQC1CFTpFyQaTg1b
...
fHZLomEYrUhsV9X+RUz6f3WnzzbFXCA=
-----END PRIVATE KEY-----
```

PEM TLS Certificate:

```
-----BEGIN CERTIFICATE-----
MIIFUjCCAzoCCQDjx3pKVc6ytTANBgkqhkiG9w0BAQsFADBtMQswCQYDVQQGEWJV
...
YfROGHT81E1aDUh17MHvRbnHI4qekw==
-----END CERTIFICATE-----
```

The PEM certificate and key must be further encoded using base64 as input to the installation screen. To encode a PEM certificate (e.g., cert.pem) to base64 on both Linux and Windows, you can use the following commands.

Linux: You can use the base64 command to read the cert.pem file, encodes it in base64, and saves the output to cert_base64.txt.

```
base64 cert.pem > cert_base64.txt
```

Windows: On Windows, you can use the certutil command to encode the cert.pem file in base64 and saves the result to cert_base64.txt.

```
certutil -encode cert.pem cert_base64.txt
```

The base64-encoded versions of the private key and certificate should look like the following:

base64-encoded TLS Private Key:

```
LS0tLS1CRUdJTiBQUk1WQVRFIEtFWs0tLS0tCk1JSUpRd01CQURBTk1na3Foa21HOXcwQkFRRUZBQVNDQ1
Mwd2dna3BBZ0VBQW9JQ0FRRE9yakUrNONQVku2QVgKV01FRFVvdnVLC01XOEJ4MTR
L...
QT1ZV1pKKz1qRkhFUTdBZXB3S1RtMEVraz0KLS0tLS1FTkQgUUFJJVkfURSBURVktLS0tLQo=
```

base64-encoded TLS Certificate:

```
LS0tLS1CRUdJTiBDRVJUSUZJQ0FURSB0tLS0tCk1JSUdGRENDQ1B5Z0F3SUJBZ01TQS9YVFJkaUpXOXMzb1
dZZ1N5a0RpZ1h5TUEwR0NTcUdTSWIZRFFQkn3VUEKTURNEN6QUpCZ05WQkFZVEFsV1RNU113RkFZRFZR
UUtFdZFNW1hRbm
...
1ND0KLS0tLS1FTkQgQ0VSVE1GSUNBVEU1LS0tLQ==
```

Prerequisite: You should prepare these base64 encoded files before the start of the installation.

- You need to have been provisioned by K2view. Provisioning includes:

- Creation of a tenant and configurations for you
- Creation of an initial K2view Admin user
- Creation of a K2view Nexus repository account
- Creation of a K2view Cloud Mailbox ID

In consultation with K2view, Docker images and their associated versions will be recommended. You will need this information to use docker pull commands to obtain images such as fabric, fabric-studio, and the k2-cloud-deployer images. These commands are described in the [Populating Your Azure Container Registry](#) section.

K2view may provide you with access to k2exchange.k2view.com optionally. Please discuss your needs with your K2view representative.

- After installing your K2view Azure Application, you will need to pause to provide K2view with information that will allow you to finalize your environment.
 - After you complete the installation and before you create a K2view project and a space, you will need to provide K2view with the location images within your container registry.
 - The list of images is those you will push as described in the [Commands to Push Containers to the Azure Container Registry](#)
 - For example, if you pushed fabric-studio:8.1.0_199 to the race8nacr.azurecr.io container registry used in this guide's examples, you will need to provide K2view with this value: race8nacr.azurecr.io/k2view/fabric-studio:8.1.0_199
 - Provide the location for every image pushed to your container registry.

2.4. Installation Checklist

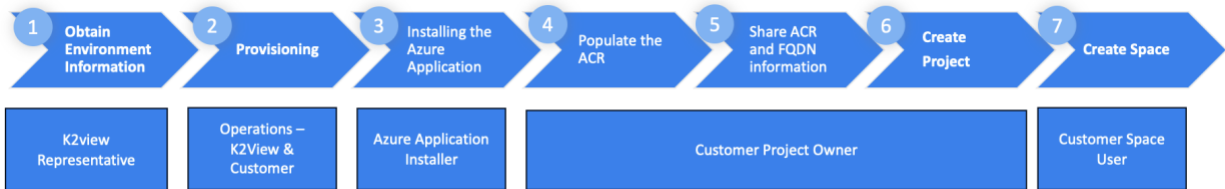
Here is a checklist to help you gather the values and content you should have before the installation:

1. Identify an individual who will install the K2view Azure Application. This individual needs to hold a Subscription Owner role.
2. Azure resources need to be registered before the installation. This individual needs the Subscription Owner role and can be the same person performing the installation.
3. Identify the Azure Region you will be installing the K2view Azure Application, and confirm that it supports multiple availability zones
4. During the installation, you will need to enter the following:
 - a. An Azure Application Resource Group name
 - b. An Azure Managed Application Resource Group name
 - c. An AKS Cluster name
 - d. An Environment Tag name
 - e. A Project Tag name
 - f. Your Domain Ingress FQDN
 - g. Obtained a Wildcard TLS PEM certificate for the Domain Ingress FQDN
 - h. Obtained the corresponding PEM private key
 - i. Base64-encode both the certificate and private key files
 - j. The K2view Cloud Mailbox ID provided to you by K2view
5. To populate your Azure Container Repository, you will need
 - a. K2view Nexus Repository Account supplied to you by K2view
 - b. Obtained the list of Docker images to pull from Nexus and push to in consultation with K2view
 - c. Docker installed on the computer used to pull and push Docker images to your ACR
6. After the image population of your ACR is complete, and before the creation of your K2view Project and your first Space, you need to provide K2view with. Once confirmed you will be able to create your K2view Project:
 - a. The FQDN of your Domain Ingress
 - b. The list of ACR locations of your Docker images
 - c. The AKS Cluster name
7. To create your K2view project, you will need:
 - a. A Git Repository URL
 - b. Git token
 - c. Git Branch / Tag
 - d. Your initial K2view Admin user

2.5. Installation Worksheet

Use this section to capture the information you need before proceeding to the installation. Please consult the [Prerequisites](#) section to understand what is needed.

At various steps during the installation, you must have obtained information from K2view and, in Step 5, provided information to K2view.



The following summarizes the information you and K2view require. Ensure that Azure resources are registered by a subscription owner before proceeding to the installation of the K2view Azure Application.

[] K2view Admin user:

Provisioned on your behalf by K2view. Please consult with your K2view representative for this information.

[] Azure Region for the K2view Azure Application Installation: _____

[] Azure Application Resource Group name: _____

[] Azure Managed Application Resource Group name: _____

[] Azure AKS Cluster name: _____

[] Azure Environment Tag name: _____

[] Azure Project Tag name: _____

[] Azure Project Tag name: _____

[] Domain Ingress FQDN: _____

[] Base64-encoded Wildcard TLS PEM certificate: This information will be in a file
Full CA chain to be included

[] Base64-encoded TLS PEM Private Key: This information will be in a file

[] K2view Cloud Mailbox ID provided to you by K2view: _____

[] K2view Nexus Repository Account supplied by K2view: _____

[] List of Docker images to pull from Nexus:
Recommended by K2view. Please consult with your K2view representative for this information.

- [] List of Docker images to pushed to your Azure Container Registry:

This list needs to be shared with K2view.

After the completion of Step 5 - the image population of your ACR is complete, and before the creation of your K2view Project and your first Space, you need to provide K2view with. Once confirmed you will be able to create your K2view Project:

- [] FQDN of your Domain Ingress:

Can be shared with K2view before the installation if you have selected the domain

- [] ACR locations of your Docker images:

The ACR created by the K2view installer generates a random name. You will need to provide this information after your installation. If you plan to use a pre-existing container registry, this information can be provided earlier.

- [] AKS Cluster name:

K2view will use the default value of k2v-aks

K2view Project Creation Step

You will need the following information to create your project.

- [] Git Repository URL:

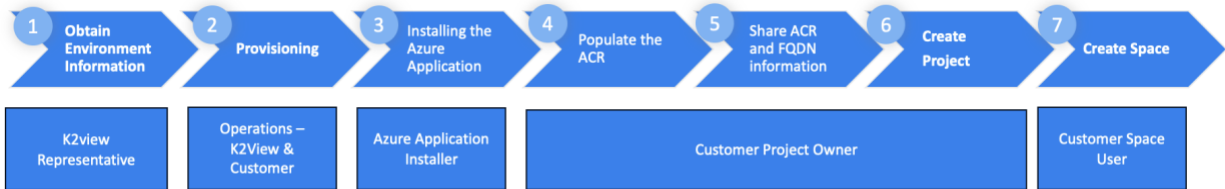
- [] Git token (do not share):

- [] Git Branch / Tag:

2.6. Post Installation

- Validation of the environment should be performed, including the DNS zone configuration. Please see the [DNS zone](#) section.
- Should SSO and IDP federation be desired, following the initial installation, additional information will be required and planned in collaboration with K2view.

3. Installing the K2view Azure Application



This section describes Step 3 – The Installation of the K2view Azure Application.

The prerequisites of this section are:

1. Identify an individual who will install the K2view Azure Application. This individual needs to hold a Subscription Owner role.
2. Azure resources need to be registered before the installation. This individual needs the Subscription Owner role and can be the same person performing the installation.
3. Identify the Azure Region where you will be installing the K2view Azure Application and confirm that it supports multiple availability zones.
4. During the installation, you will need to enter the following:
 - a. An Azure Application Resource Group name
 - b. An Azure Managed Application Resource Group name
 - c. An AKS Cluster name
 - d. An Environment Tag name
 - e. A Project Tag name
 - f. Your Domain Ingress FQDN
 - g. Obtained a Wildcard TLS PEM certificate for the Domain Ingress FQDN
 - h. Obtained the corresponding PEM private key
 - i. Base64 encoded both versions of the certificate and private key files
 - j. The K2view Cloud Mailbox ID provided to you by K2view

Before you begin the installation process

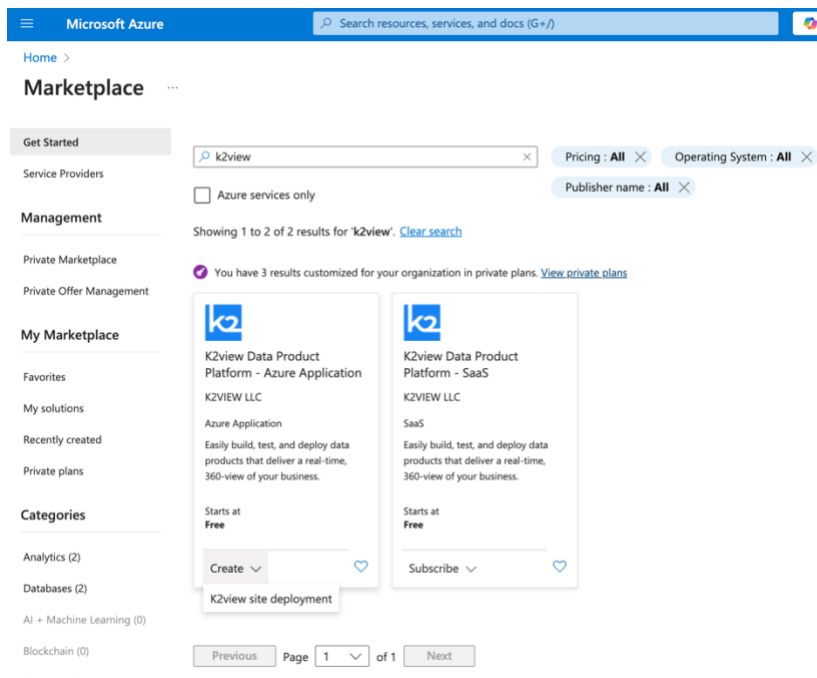
Azure resources need to be registered before the installation as described in the [Azure Resource Registration](#) section.

Proceeding to the installation

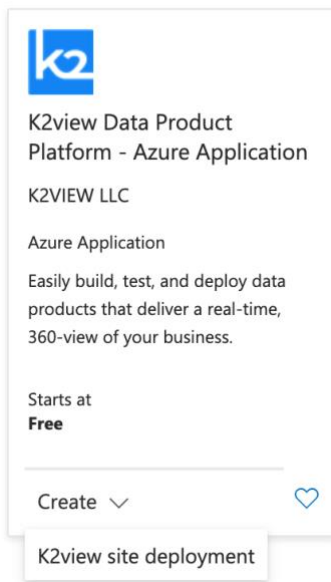
The Azure Application installer is available and found on the Azure Marketplace. Search for “K2view” in the marketplace as shown below. For the purpose of this installation you will use the “K2view Data Product Platform – Azure Application”.

This application will fully install Fabric services within a resource group of your Azure subscription. The installation process is fully automated and requires only minimal input provided in the [Installation Checklist](#) section. Please review the [Planning](#) in its entirety to understand the constraints, prerequisites, and the meaning of the terms used.

Search for K2view



Select the Azure Application listing and press Create.



The installation process will guide you through several screens. You must select the appropriate Azure subscription, a resource group that can be a preexisting one or one you create, and the region where the Fabric services will be installed. Additionally, you must provide the name of the Azure-managed application and its associated resource group.

Please consult the [Constraints](#) and [Prerequisites](#) section to ensure you have been assigned the Subscription Owner role and identified an Azure region that supports multiple availability zones.

First Page of the Installation Wizard

Microsoft Azure

Search resources, services, and docs (G+I)

Copilot

luc.clement@k2view.com
K2VIEW (P&D)

Home > Marketplace >

Create K2view site deployment

Basics

K2view Installation settings

K2view Application settings

Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

K2view PAYG Marketing

Resource group *

Create new

Instance details

Region

Loading...

Managed Application Details

Provide a name for your managed application, and its managed resource group. Your application's managed resource group holds all the resources that are required by the managed application.

Application Name *

Managed Resource Group *

mrg-k2view-azure-application-20241024140403

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot



luc.clement@k2view.com
K2VIEW (R&D)

Home > Marketplace >

Create K2view site deployment

Basics

K2view Installation settings

K2view Application settings

Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

K2view PAYG Marketing

Resource group *

acm-k2view-rg

Create new

Instance details

Region *

East US 2

Managed Application Details

Provide a name for your managed application, and its managed resource group. Your application's managed resource group holds all the resources that are required by the managed application.

Application Name *

k2view360

Managed Resource Group *

mrg-k2view-azure-application

Previous

Next

Review + create

Give feedback

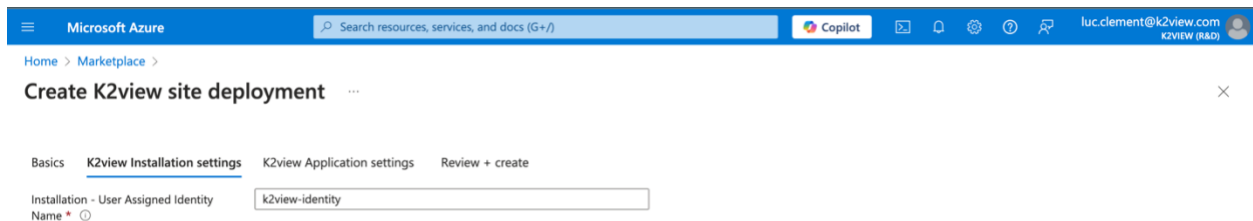
The installation uses two resource groups.

- **Azure Application Resource Group.** This is shown on the page below “Subscription”. You can choose an existing resource group or create a new one. You should select a representative name. In the example demonstrated in this guide we use “acm-k2view-rg”. Select something that has significance to your project.
- **Azure Managed Application Resource Group.** This is shown on the page as “Managed Resource Group”. K2view resources will be created in a separate resource group, the Azure Managed Application resource group. This resource group contains K2view infrastructure components created during the installation. A name will be suggested for you. You can use that name or one that has significance to you. In our example, we used **mrg-k2view-azure-application**.

Press Next at the bottom of the page.

Second Page of the Installation Wizard

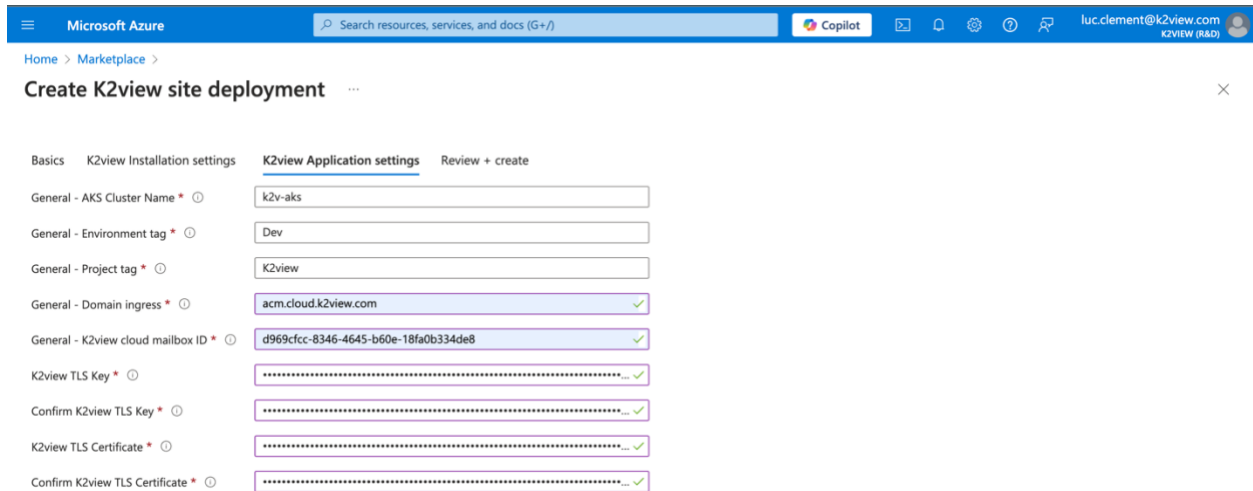
On this screen you set the identity used for the installation. Unless you have specific needs to do otherwise, we recommend that you use the default of “k2view-identity”



Press Next at the bottom of the page.

Third Page of the Installation Wizard

The third step of the process is where you provide the details you will find in the [Installation Checklist](#).



Microsoft Azure | Search resources, services, and docs (G+/I) | Copilot | luc.clement@k2view.com | K2VIEW (RAO)

Home > Marketplace > Create K2view site deployment

Basics | K2view Installation settings | **K2view Application settings** | Review + create

General - AKS Cluster Name *

General - Environment tag *

General - Project tag *

General - Domain ingress *

General - K2view cloud mailbox ID *

K2view TLS Key *

Confirm K2view TLS Key *

K2view TLS Certificate *

Confirm K2view TLS Certificate *

Please enter

- **Cluster Name:** This is an Azure Kubernetes Service (AKS) cluster name of the Kubernetes cluster to be created. The default, if used, is k2v-aks.
- **Environment Tag:** This tag describes the project's environment. Typical values could be Dev, UAT, and Prod. Azure cost management modules will use this tag to help you categorize costs.
- **Project Tag:** This tag describes the functional purpose of the project. You might consider tag names such as K2view TDM, K2view Masking, and K2view 360. Azure cost management modules will use this tag to help you categorize costs.
- **Domain Ingress:** This is the FQDN of the Nginx Ingress Controller for which a wildcard certificate needs to be created. In our example, we used acm.cloud.k2view.com, a domain owned by K2view. You will need to select your own domain and one for which you can create TLS certificates (PEM encoded).
- **TK2view Cloud Mailbox ID** provided to you by K2view
- **The base64-encoded values of your PEM TLS certificate and its associated private key**

Your network team must create a wildcard TLS certificate for the domain you selected as your Domain Ingress. The full certificate authority chain needs to be present. During the installation, you will need to paste the base64-encoded values of your TLS certificate and its associated private key.

- **TLS Private:** the base64-encoded private key associated with the TLS certificate
- **TLS Certificate:** a base64-encoded wildcard certificate. For example, for the domain ingress of acm.cloud.k2view.com you need to create a wildcard certificate for *.acm.cloud.k2view.com for hosts of this domain.

What you paste in should look like the following:

base64-encoded TLS Private Key:

```
LS0tLS1CRUdJTiBQUk1wQVRFIEtFWS0tLS0tCk1JSUpRd01CQURBTk1na3Foa21HOXcwQkFRRUZBQVNDQ1  
Mwd2dna3BBZ0VBQW9JQ0FRRE9yakUrNONQVku2QVgKV01FRFVvdnVlc01XOEJ4MTR  
L...  
QT1ZV1pKKz1qRkhFUTdBZXB3S1RtMEVraz0KLS0tLS1FTkQgUUFJJVkfURSBRLRVktLS0tLQo=
```

base64-encoded TLS Certificate:

```
LS0tLS1CRUdJTiBDRVJUSUZJQ0FURSB0tLS0tCk1JSUdGRENDQ1B5Z0F3SUJBZ01TQS9YVFJkaUpXOXMzb1  
dZZlN5aORpZ1h5TUEwR0NTcUdTSWlZRFFFQkN3VUEKTURNEN6QUpCZ05WQkFZVEFsV1RNU113RkFZRFRZ  
UUtFdzFNW1hRbm  
...  
1ND0KLS0tLS1FTkQgQ0VSVE1GSUNBVEUtLS0tLQ==
```

Note: The PEM certificate and key must be further encoded using base64 as input to the installation screen. To encode a PEM certificate (e.g., cert.pem) to base64 on both Linux and Windows, you can use the following commands.

Linux: You can use the base64 command to read the cert.pem file, encodes it in base64, and saves the output to cert_base64.txt.

```
base64 cert.pem > cert_base64.txt
```

Windows: On Windows, you can use the certutil command to encode the cert.pem file in base64 and saves the result to cert_base64.txt.

```
certutil -encode cert.pem cert_base64.txt
```

Prerequisite: You should prepare these base64 encoded files before the start of the installation.

Press Next at the bottom of the page.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot



luc.clement@k2view.com
K2VIEW (P&O)

Home > Marketplace >

Create K2view site deployment

activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Name

Luc Clement

Preferred e-mail address

luc.clement@k2view.com

Preferred phone number

Basics

Subscription

K2view PAYG Marketing

Resource group

acm-k2view-rg

Region

East US 2

Application Name

k2view360

Managed Resource Group Name

mrg-k2view-azure-application

K2view Installation settings

Installation - User Assigned Identity Na... k2view-identity

K2view Application settings

General - AKS Cluster Name

k2v-aks

General - Environment tag

Dev

General - Project tag

K2view

General - Domain ingress

acm.cloud.k2view.com

General - K2view cloud mailbox ID

d969cfcc-8346-4645-b60e-18fa0b334de8

K2view TLS Key

.....

K2view TLS Certificate

.....

Previous

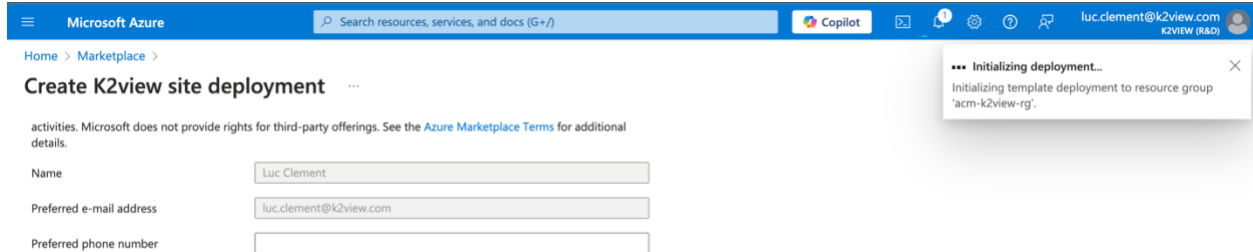
Next

Create

Give feedback

The Installation Process Begins

It will take approximately 15 minutes to create all resources in the **Azure Application Resource Group** And **Azure Managed Application Resource Group** you selected on the first page of the installation wizard.



Microsoft Azure

Search resources, services, and docs (G+)

Copilot

luc.clement@k2view.com K2VIEW (RAD)

Home > Marketplace >

Create K2view site deployment

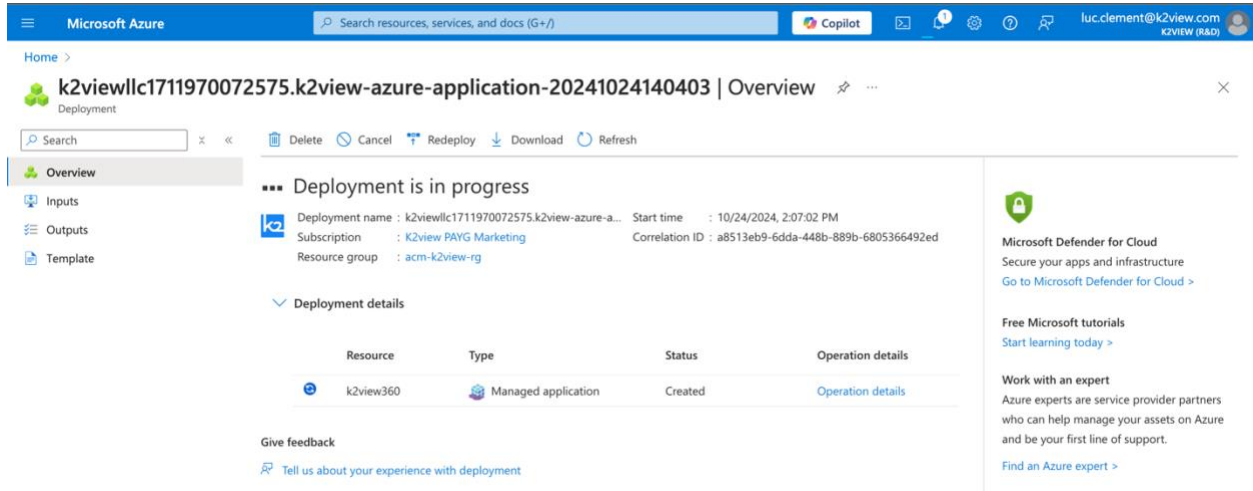
activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Name:

Preferred e-mail address:

Preferred phone number:

*** Initializing deployment...
Initializing template deployment to resource group 'acm-k2view-rg'.



Microsoft Azure

Search resources, services, and docs (G+)

Copilot

luc.clement@k2view.com K2VIEW (RAD)

Home >

k2viewllc1711970072575.k2view-azure-application-20241024140403 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name : k2viewllc1711970072575.k2view-azure-a... Start time : 10/24/2024, 2:07:02 PM
Subscription : K2view PAYG Marketing Correlation ID : a8513eb9-6dda-448b-889b-6805366492ed
Resource group : acm-k2view-rg

Deployment details

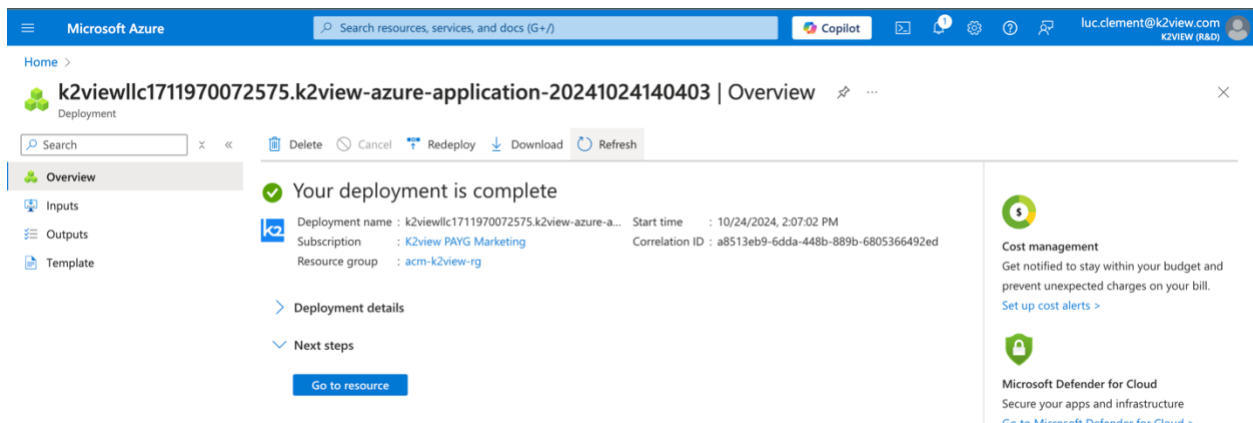
Resource	Type	Status	Operation details
k2view360	Managed application	Created	Operation details

Give feedback
[Tell us about your experience with deployment.](#)

Microsoft Defender for Cloud
Secure your apps and infrastructure
[Go to Microsoft Defender for Cloud >](#)

Free Microsoft tutorials
[Start learning today >](#)

Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
[Find an Azure expert >](#)



Microsoft Azure

Search resources, services, and docs (G+)

Copilot

luc.clement@k2view.com K2VIEW (RAD)

Home >

k2viewllc1711970072575.k2view-azure-application-20241024140403 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name : k2viewllc1711970072575.k2view-azure-a... Start time : 10/24/2024, 2:07:02 PM
Subscription : K2view PAYG Marketing Correlation ID : a8513eb9-6dda-448b-889b-6805366492ed
Resource group : acm-k2view-rg

Deployment details

Next steps

[Go to resource](#)

Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

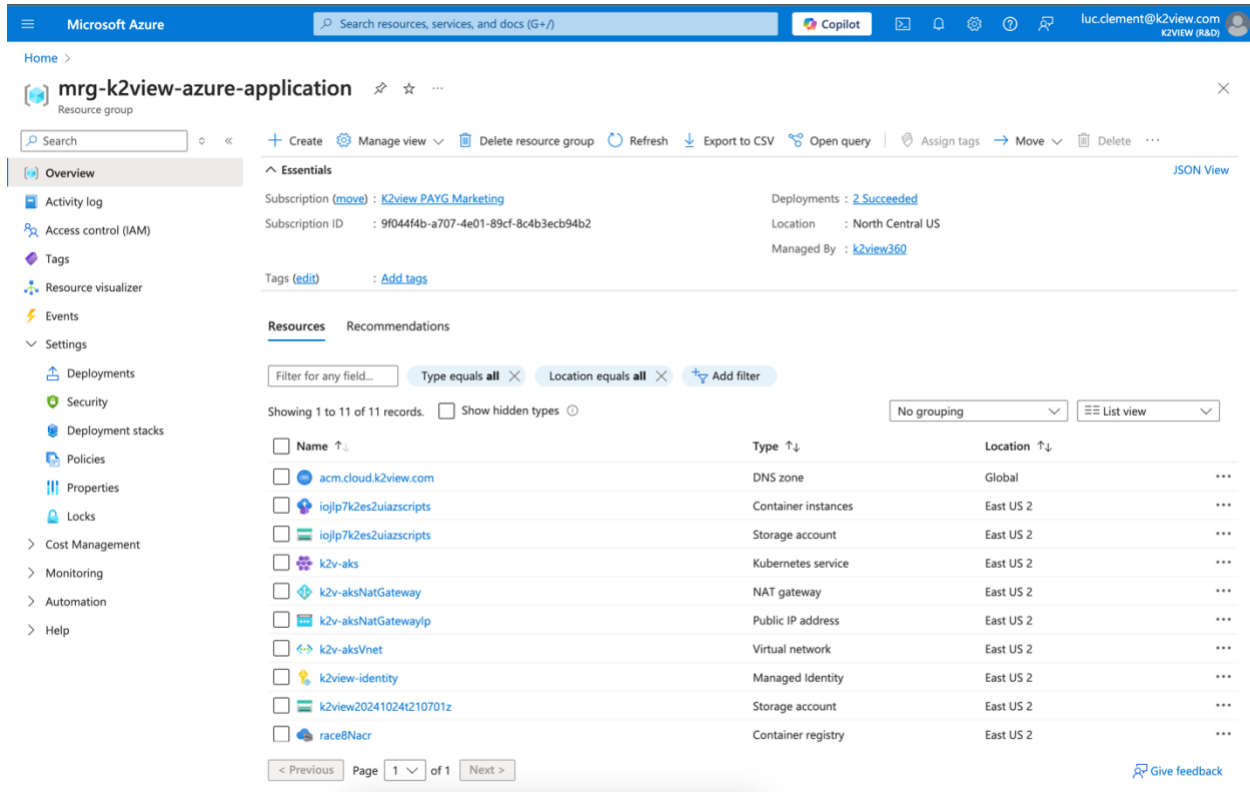
Microsoft Defender for Cloud
Secure your apps and infrastructure
[Go to Microsoft Defender for Cloud >](#)

The Azure Application installation has been completed.

You will find the following created resources by navigating to the newly created resource group.

A Terraform module is used behind the scenes to take these input parameters to deploy K2View Fabric services in your environment. For more information on the details and configurations performed by the Azure Application installer, you can review the Terraform module available at:

<https://github.com/k2view/blueprints/tree/main/Azure/terraform/AKS>.

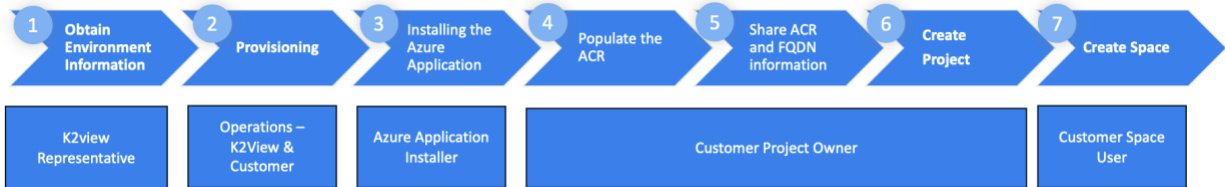


The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The main content area displays the 'mrg-k2view-azure-application' resource group. The 'Resources' tab is selected, showing a list of 11 resources. The resources are listed in a table with columns for Name, Type, and Location. The resources include:

Name	Type	Location
acm.cloud.k2view.com	DNS zone	Global
iojlp7k2es2uiazscripts	Container instances	East US 2
iojlp7k2es2uiazscripts	Storage account	East US 2
k2v-aks	Kubernetes service	East US 2
k2v-aksNatGateway	NAT gateway	East US 2
k2v-aksNatGatewayip	Public IP address	East US 2
k2v-aksVnet	Virtual network	East US 2
k2view-identity	Managed Identity	East US 2
k2view20241024t210701z	Storage account	East US 2
race8Nacr	Container registry	East US 2

You are now ready to proceed to the next step: populate the Azure Container Registry created for you with K2view service Docker images.

4. Populating Your Azure Container Registry



To populate your Azure Container Repository, you will need

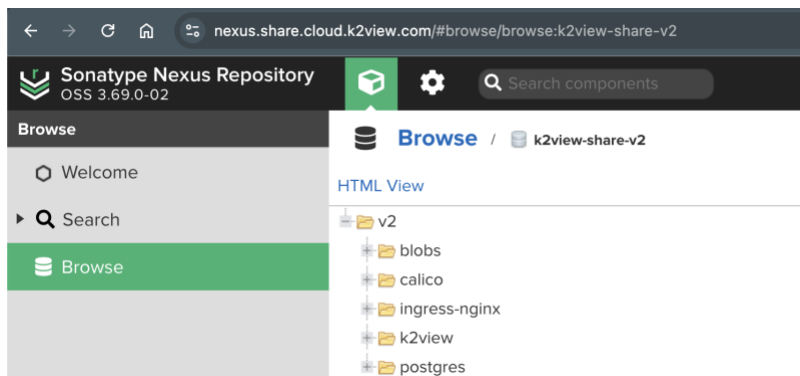
1. K2view Nexus Repository Account supplied to you by K2view
2. Obtained the list of Docker images to pull from Nexus and push to in consultation with K2view
3. Docker installed on the computer used to pull and push Docker images to your ACR

The [Planning](#) section describes what is necessary to be ready to perform this step.

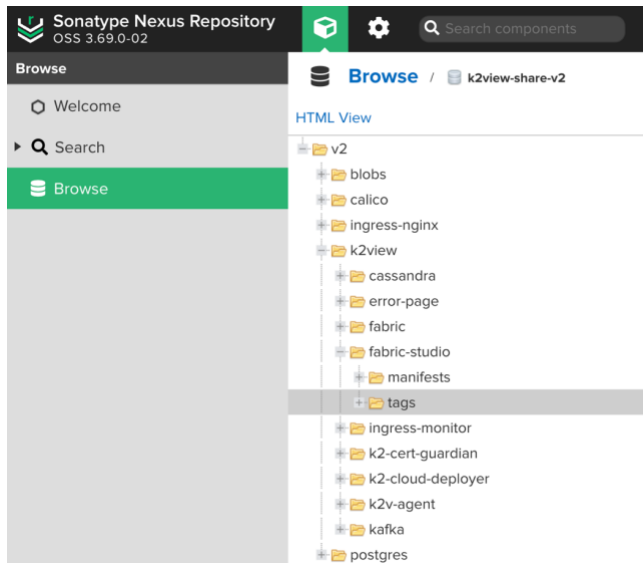
4.1. Getting Acquainted with the K2view Nexus Repository

To acquaint yourself of the K2view Docker images available to you can you login to the K2view Nexus web application. Here you can find the versions available to you. Please consult with your K2view representative which images and versions you should use. We recommend using the latest version of the K2view Docker images.

Login and navigate to <https://nexus.share.cloud.k2view.com/#browse/browse:k2view-share-v2> to view available images



For example, you can find the latest version of K2view's Fabric Studio by expanding the tags folder.



4.2. Commands to Pull Containers from the K2view Nexus Repository

The list of K2view service Docker images will be discussed and recommended to you in consultation with your K2view representative.

For the purpose of these instructions we will “pull” from the K2view Nexus repository three images of a specific version: fabric-studio, fabric, and the k2-cloud-deployer Docker images.

Docker tools need to be installed on the computer you are performing this installation. You can download these from <https://docker.com>.

First Login

Command:

```
docker login -u [your K2view Nexus username] https://docker.share.cloud.k2view.com
```

Second, pull the required images. In this example we pull three.

Commands: In succession, please run the following three commands. Keep in mind that the version and docker images needed may differ.

```
docker pull docker.share.cloud.k2view.com/k2view/fabric-studio:8.1.0_199
```

```
docker pull docker.share.cloud.k2view.com/k2view/fabric:8.1.0_199
```

```
docker pull docker.share.cloud.k2view.com/k2view/k2-cloud-deployer:1.8.0
```

4.3. Commands to Push Containers to the Azure Container Registry

Now that you have “pulled” images from the K2view Nexus repository, you are now ready to “push” these to the Azure Container Registry (ACR) created for you. If you wish to use a preexisting ACR or OCI-compliant container registry, please also consult the [Selecting a Different Azure Container Registry or OCI-Compliant Registry](#) section.

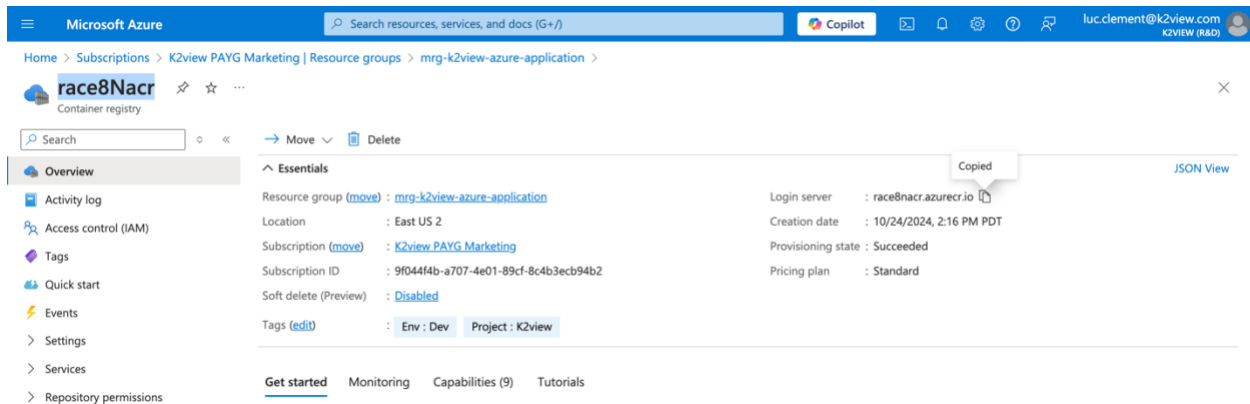
First, obtain the credentials required to perform a Docker Push to your ACR

Installing the K2view Azure Application created an ACR within the Azure Managed Application Resource Group. You can obtain the credential associated with it by navigating you the resource group and selecting the ACR created for you.

In this example, the Managed Application Resource Group is named mgr-k2view-azure-application, and the created ACR is named rac8Nacr (a random name).

When you navigate to this ACR location, please

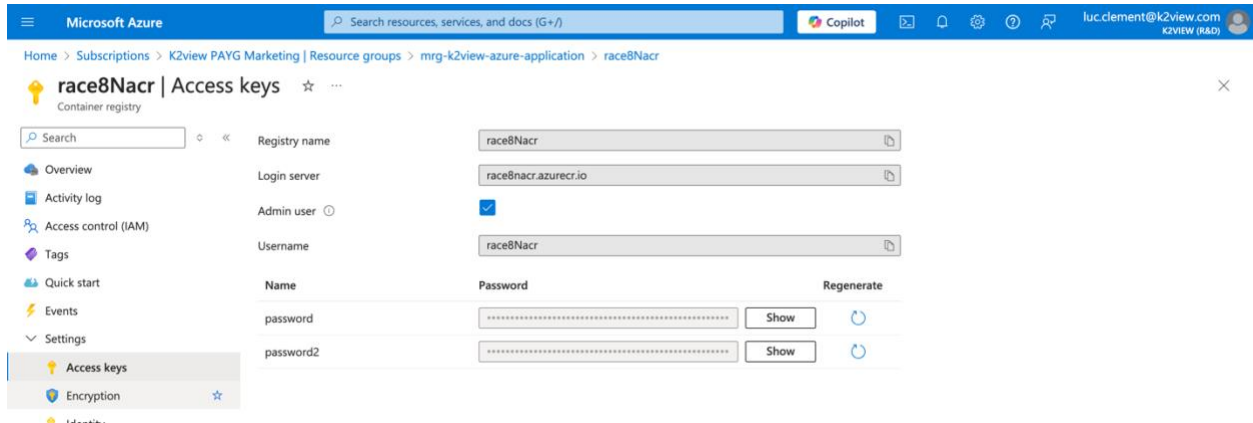
1. Get the URL of the login server, which is shown here. Copy the URL which will be required for your Docker login and push commands.



The screenshot shows the Microsoft Azure portal interface. The breadcrumb navigation at the top indicates the path: Home > Subscriptions > K2view PAYG Marketing | Resource groups > mgr-k2view-azure-application > race8Nacr. The 'race8Nacr' resource is identified as a 'Container registry'. On the left, a sidebar lists navigation options: Overview, Activity log, Access control (IAM), Tags, Quick start, Events, Settings, Services, and Repository permissions. The main content area displays the 'Overview' page for the ACR. It includes a search bar, action buttons for 'Move' and 'Delete', and a 'JSON View' link. The 'Essentials' section provides key details: Resource group (mgr-k2view-azure-application), Location (East US 2), Subscription (K2view PAYG Marketing), Subscription ID (9f044f4b-a707-4e01-89cf-8c4b3ecb94b2), Soft delete (Preview) (Disabled), and Tags (Env: Dev, Project: K2view). A 'Login server' field shows 'race8nacr.azurecr.io' with a copy icon and a 'Copied' tooltip. Other fields include Creation date (10/24/2024, 2:16 PM PDT), Provisioning state (Succeeded), and Pricing plan (Standard). At the bottom, there are tabs for 'Get started', 'Monitoring', 'Capabilities (9)', and 'Tutorials'.

2. Select Access Keys in the left navigation menu under Settings to obtain your credentials. From this page:
 - a. Copy the username
 - b. Use the Show button to obtain the password, and copy this password

This information will be required to perform the login to your ACR using a Docker command



Second, perform the Docker Push to your ACR

Use the Docker login command to login to your ACR

```
docker login -u [your ACR username] [the ACR Login server]
```

In our example we used: `docker login -u race8Nacr race8nacr.azurecr.io`

Third, tag each image you pulled from the K2view Nexus

Use of tags is for version control. Tagging Docker images allows you to maintain different versions of the same application, making it easier to roll back to a previous version if needed. This can be especially useful in cases where a new release introduces a bug or breaks compatibility with other components.

In our example, we used the following three commands, respectively, for the fabric-studio, fabric, and k2-cloud-deployer Docker images. Note that the highlighted **race8nacr** value is the name used in our example, you will need to use the one associated with your ACR.

```
docker tag docker.share.cloud.k2view.com/k2view/fabric-studio:8.1.0_199  
race8nacr.azurecr.io/k2view/fabric-studio:8.1.0_199
```

```
docker tag docker.share.cloud.k2view.com/k2view/fabric:8.1.0_199  
race8nacr.azurecr.io/k2view/fabric:8.1.0_199
```

```
docker tag docker.share.cloud.k2view.com/k2view/k2-cloud-deployer:8.1.0_199  
race8nacr.azurecr.io/k2view/k2-cloud-deployer:8.1.0_199
```


Fourth, push images to your ACR

Next, you need to push the Docker images you pulled from the K2view Nexus to your ACR using the command: `docker push [image location]`.

In our example, we used:

```
docker push race8nacr.azurecr.io/k2view/fabric-studio:8.1.0_199
```

```
docker push race8nacr.azurecr.io/k2view/fabric:8.1.0_199
```

```
docker push race8nacr.azurecr.io/k2view/k2-cloud-deployer:1.8.0
```

Fifth, please provide the locations you used to K2view

K2view needs these ACR locations of your images to finalize your site's configuration. Please note that you will share these locations with K2view as described in the [Sharing Information with K2view](#) section.

4.4. Selecting a Different Azure Container Registry or OCI-Compliant Registry

You must follow similar steps if you choose to use a pre-existing ACR or an OCI-compliant registry. You will also need to provide K2view with the locations of your registry.

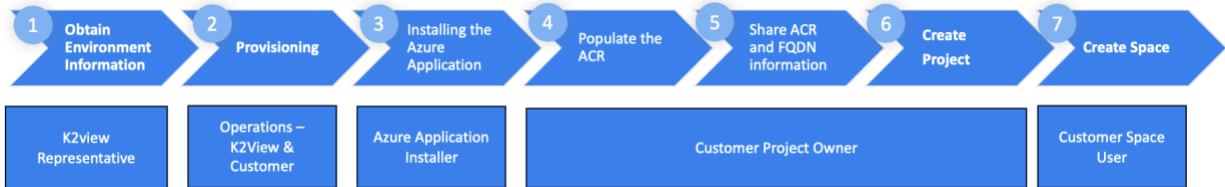
4.5. Sharing Information with K2view

The Customer Project Owner must share the following information to complete the configuration for the tenant's creation of Projects and Spaces.

- The fully qualified domain name (FQDN) used for your Domain Ingress.
 - For example, `acm.cloud.k2view.com` was used. You need to use a domain your organization owns.
- The list of Azure Container Registry (ACR) locations of your Docker images
 - E.g., the locations shown in the fourth step of the [Commands to Push Containers to the Azure Container Registry](#) section.
- The AKS Cluster name

This information may have been identified during pre-planning, but the specific information should be confirmed to avoid misconfiguration.

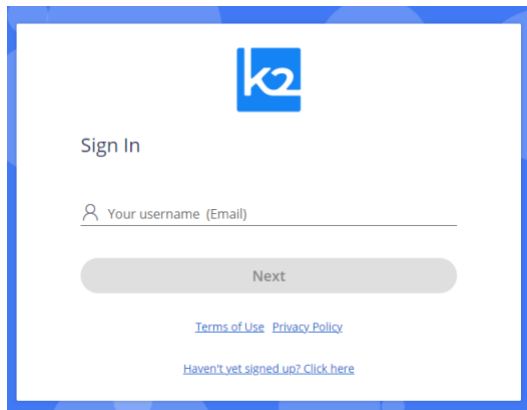
5. Creating Your Fabric Project



5.1. Sign In to K2cloud

To sign in, access cloud.k2view.com using Chrome, Microsoft Edge, or Firefox web browsers.

When prompted to sign in, enter your email address. This account needs to have been provisioned for you by K2view. Please refer to the [Prerequisites](#) section.



If provisioned by K2view, you will receive an access code via email you will then enter in the prompt window that is shown when you press Next.

You will be directed to the Spaces page, where you will eventually create spaces. First, you need to create a project.

5.2. Create the Project

You create a project by selecting the Projects top-level navigation tab and pressing “Create Project”. A project requires a name and an associated Git repository. You must provide Git-related information to create a project. To learn more about Git and how to obtain the Repository URL you will use, a branch, and your token, please review the [GitHub](#) section.

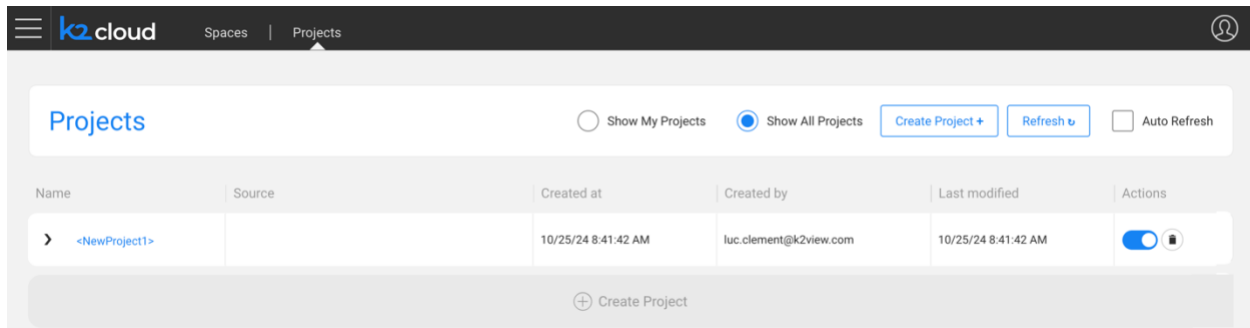
The Fabric spaces you implement are based on a “Space Profile”. You can configure more than one Space Profile per Project. If you need assistance, a K2view representative will assist you. Some of this information will have been shared with you during the [Planning](#) phase.

To create a Space Profile, you need to:

- Give it a name that will have significance to your users
- Select an Image Profile identified in consultation with K2view, and
- Define the site where this space will operate within. K2view will help you define this list.

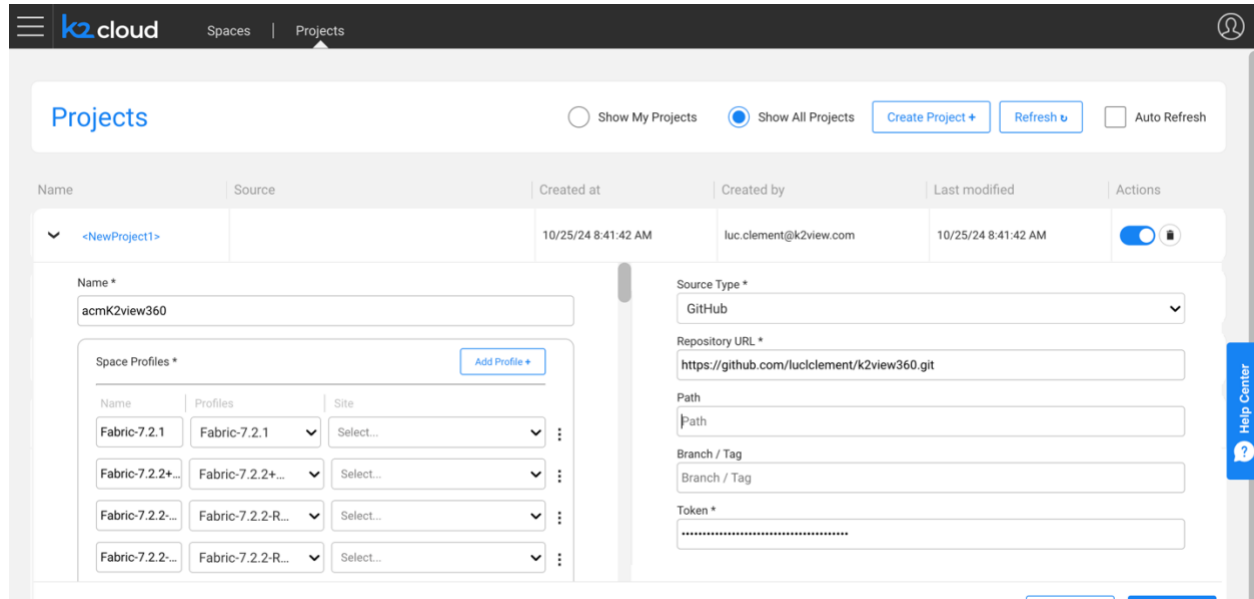
Creating a Project

After pressing Create Project, an empty project will be created.



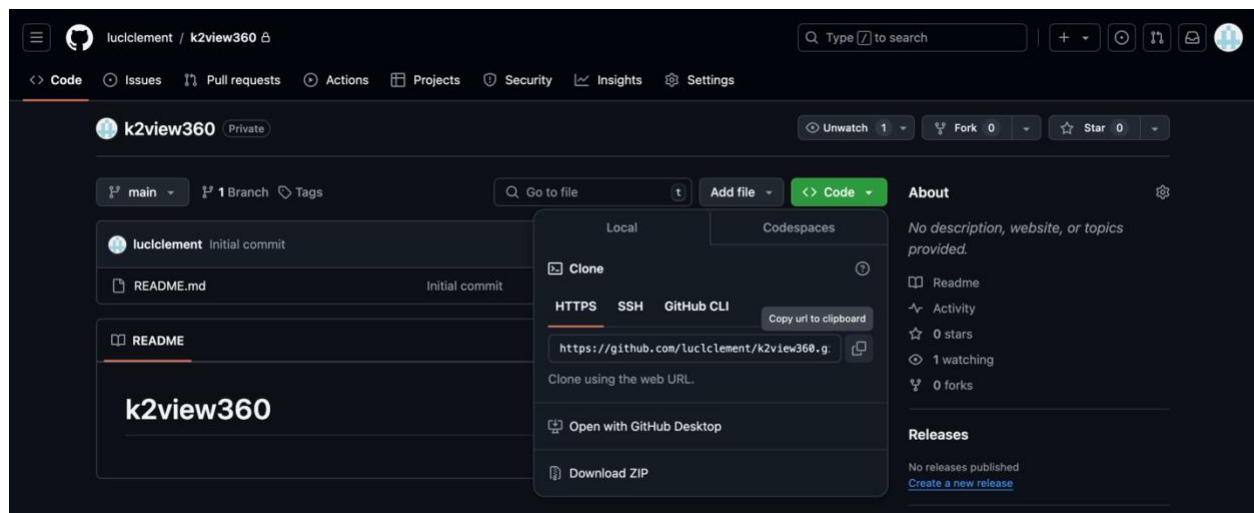
Provide the project with a name and select your space profiles. Again, K2view will have provided this.

You also need to provide the details of your Git source control system described in the [GitHub](#) section.

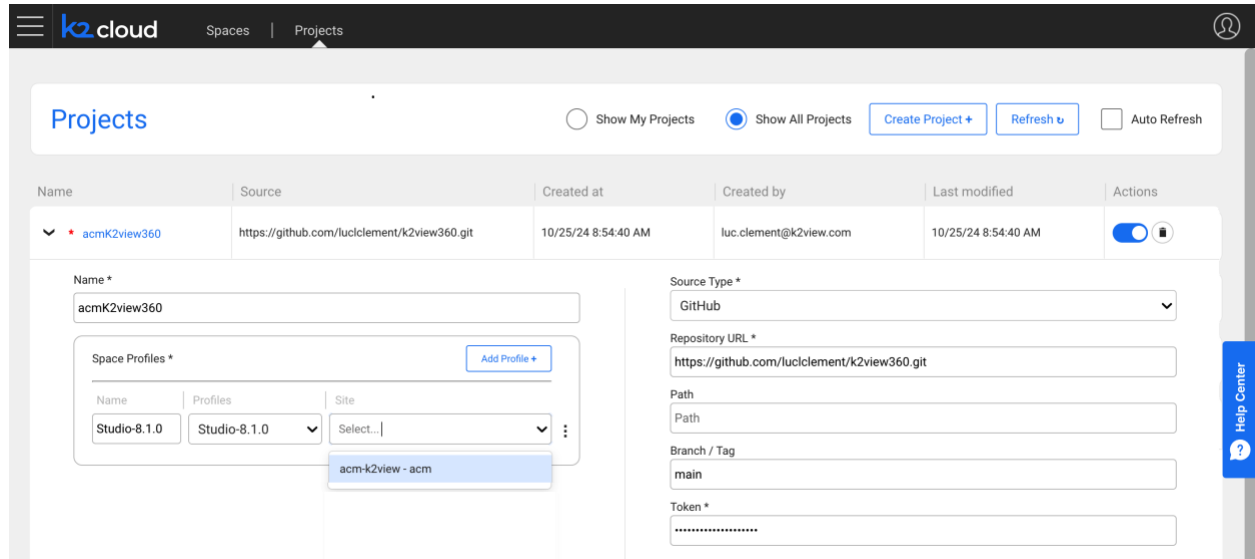


The screenshot shows the 'Projects' configuration page in the K2view cloud interface. At the top, there are tabs for 'Spaces' and 'Projects', with 'Projects' selected. Below the tabs, there are buttons for 'Show My Projects', 'Show All Projects' (selected), 'Create Project +', 'Refresh', and 'Auto Refresh'. A table lists projects with columns: Name, Source, Created at, Created by, Last modified, and Actions. The first project is '<NewProject1>'. Below the table, there is a form to create a new project. The 'Name' field is 'acmK2view360'. The 'Source Type' is 'GitHub'. The 'Repository URL' is 'https://github.com/luccllement/k2view360.git'. The 'Path' field is empty. The 'Branch / Tag' field is empty. The 'Token' field is empty. There is a section for 'Space Profiles' with a table of profiles and a 'Select...' dropdown for each. The table has columns: Name, Profiles, and Site. The profiles listed are 'Fabric-7.2.1', 'Fabric-7.2.2+', 'Fabric-7.2.2...', 'Fabric-7.2.2-R...', and 'Fabric-7.2.2-R...'. The 'Site' column has a 'Select...' dropdown for each profile.

For example, you can obtain the URL from your Git repository, as shown here in our example.



And, finally, please select the site created from the site's dropdown.



The screenshot shows the K2view cloud interface. At the top, there's a navigation bar with 'Spaces' and 'Projects' tabs. The 'Projects' tab is active, showing a table of projects. Below the table, there's a form for creating a new project. The form includes fields for Name, Source Type, Repository URL, Path, Branch / Tag, and Token. A dropdown menu for 'Site' is open, showing a list of sites, with 'acm-k2view - acm' selected.

Name	Source	Created at	Created by	Last modified	Actions
acmK2view360	https://github.com/luccllement/k2view360.git	10/25/24 8:54:40 AM	luc.clement@k2view.com	10/25/24 8:54:40 AM	 

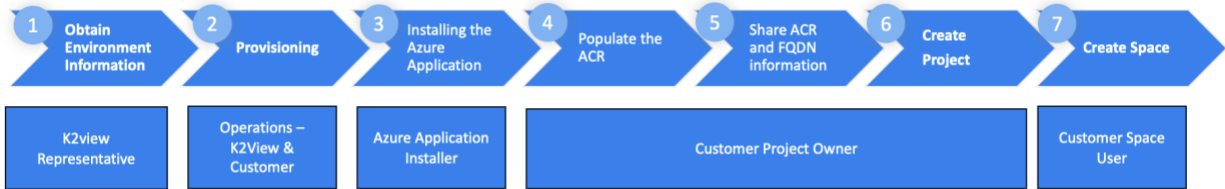
Form Fields:

- Name *: acmK2view360
- Source Type *: GitHub
- Repository URL *: https://github.com/luccllement/k2view360.git
- Path: Path
- Branch / Tag: main
- Token *:
- Space Profiles *:

Name	Profiles	Site
Studio-8.1.0	Studio-8.1.0	Select...

 (Dropdown menu open showing: acm-k2view - acm)

6. Creating Your First Fabric Space



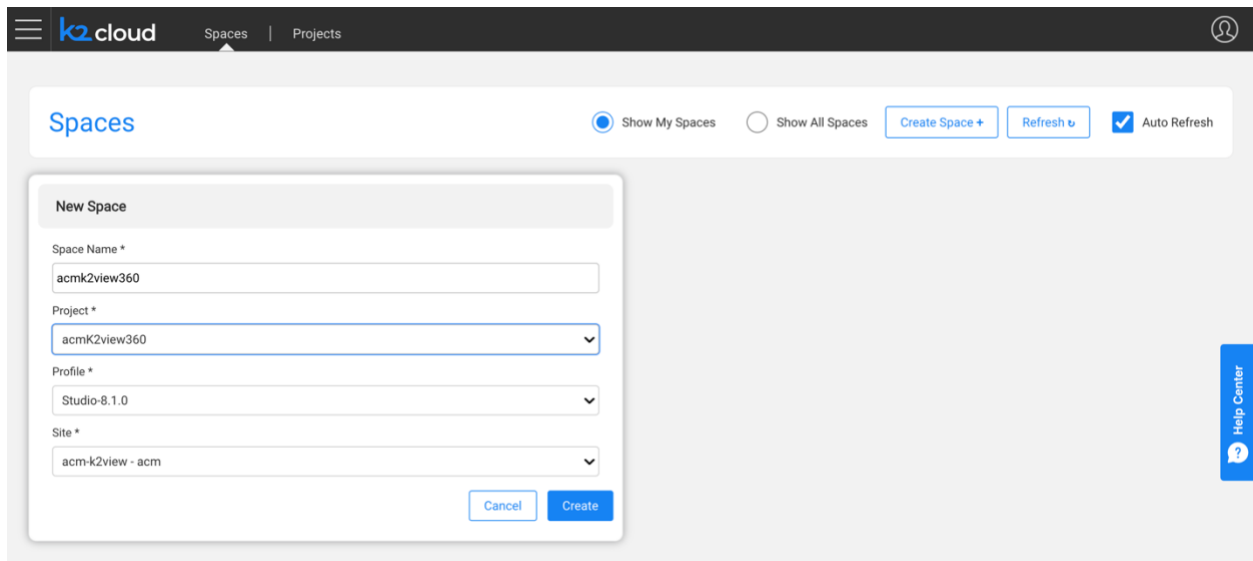
After your Project is created, you can create a Fabric Space. Here, we show an example of an existing space and a new space being defined.

To create a space

- Give it a name,
- Select the project you created,
- Select your intended profile, one recommended by K2view given your intended purpose, and
- Select site.

If you need to install a K2view extension such as TDM, you must go to your space and install the extension from K2exchange.

When ready, press Create to trigger the space's creation.

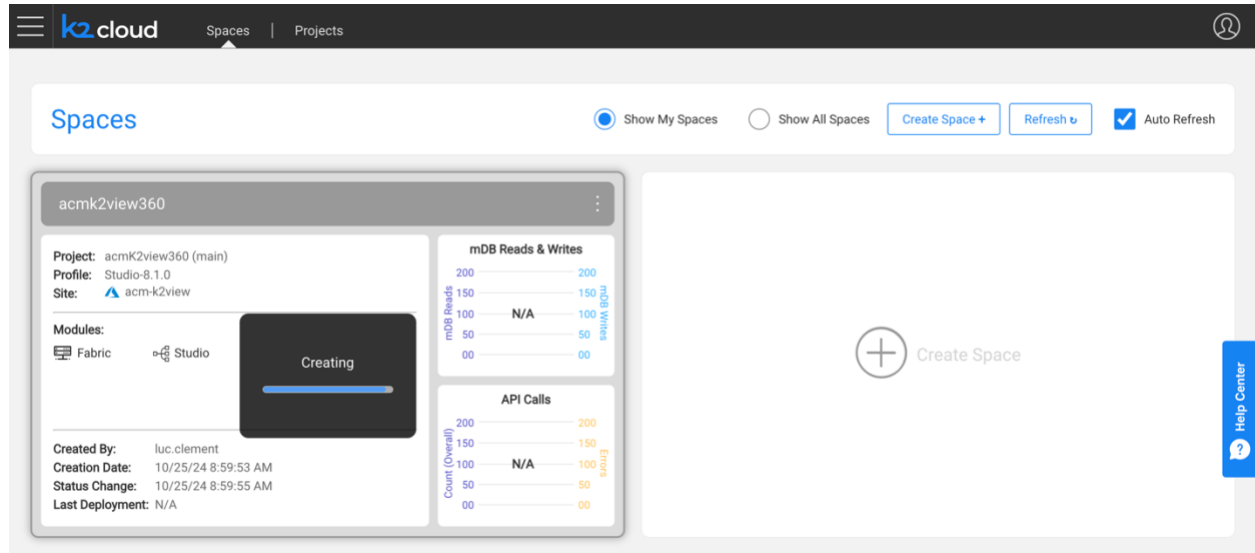


The screenshot shows the K2cloud interface with the 'Spaces' tab selected. A 'New Space' modal is open, allowing the user to create a new space. The modal contains the following fields:

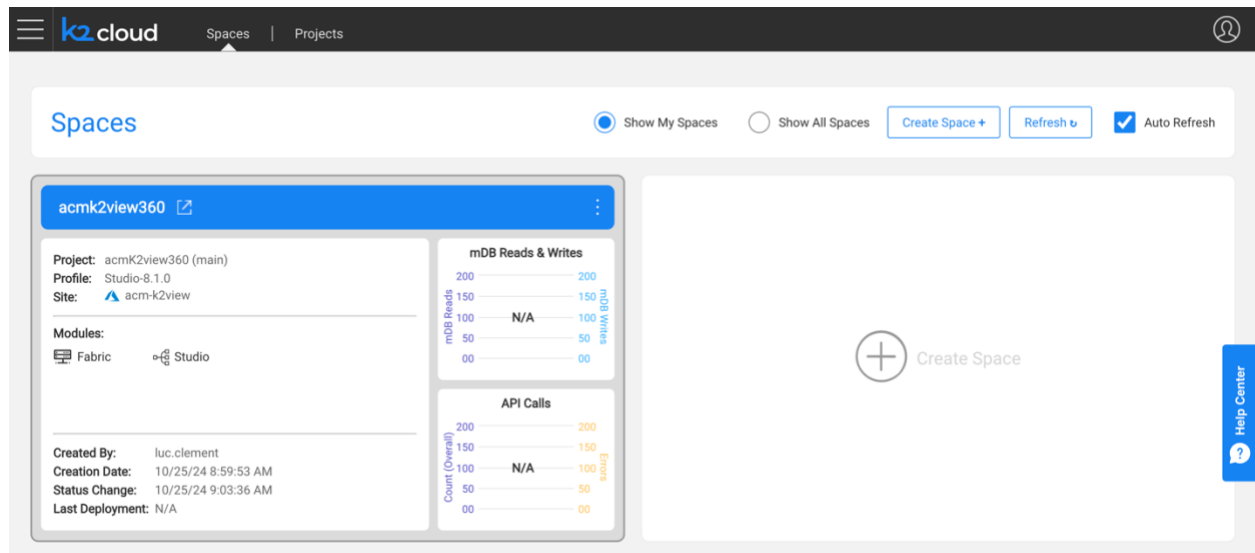
- Space Name ***: acmk2view360
- Project ***: acmk2view360 (selected from a dropdown)
- Profile ***: Studio-8.1.0 (selected from a dropdown)
- Site ***: acm-k2view - acm (selected from a dropdown)

At the bottom of the modal are 'Cancel' and 'Create' buttons. In the background, the 'Spaces' page shows a list of spaces and a 'Create Space +' button. The top navigation bar includes the K2cloud logo and a user profile icon.

The Space creation process begins and may take a few minutes to complete.



Once complete your site is displayed to you.



You have successfully created your Project and Space and are ready to begin working with K2view.

7. Validation

You are now ready to proceed with validating your space. Doing so is performed by selecting the top right arrow icon (see next section).

This will cause the browser to connect to the Space you created over HTTPS to the environment you created. The TLS certificate you assigned will be used for this connection.

DNS

During the installation, a DNS zone was created. This zone may take time to propagate into your environment and may not be available until it does.

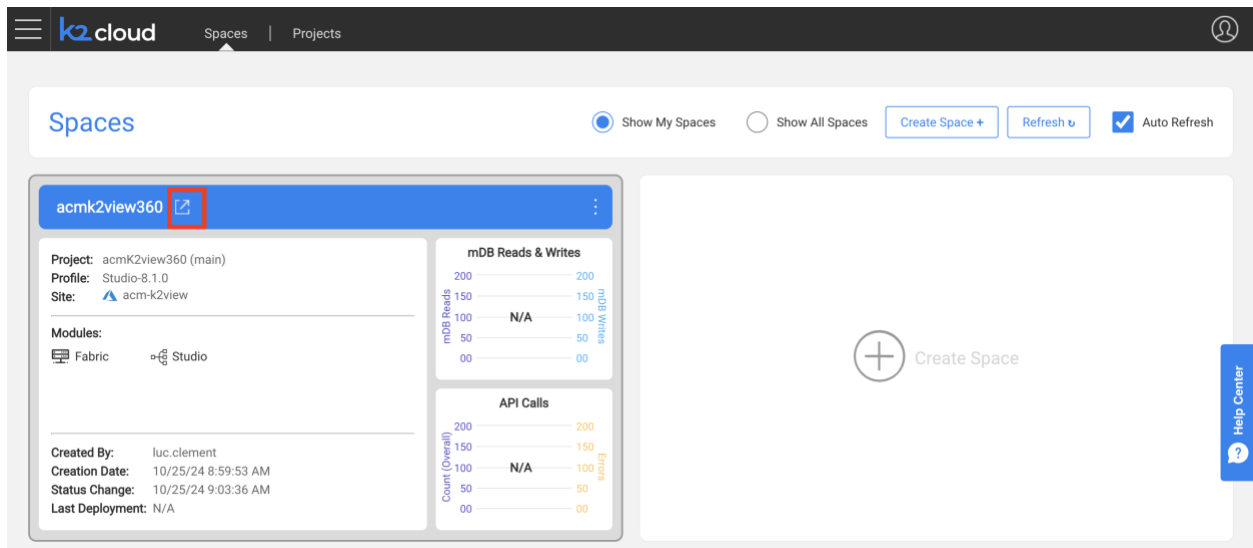
You should note that when creating a namespace, its name is associated as a subdomain to this domain name in the Ingress controller. For example, if the domain is "k2dev.company.com" and a created namespace is "test", then the URL of this namespace that users shall access will be "test.k2dev.company.com".

In our example, we will connect to a space accessible at acmk2view360-k2v-rnd.acm.cloud.k2view.com, a publicly resolvable domain.

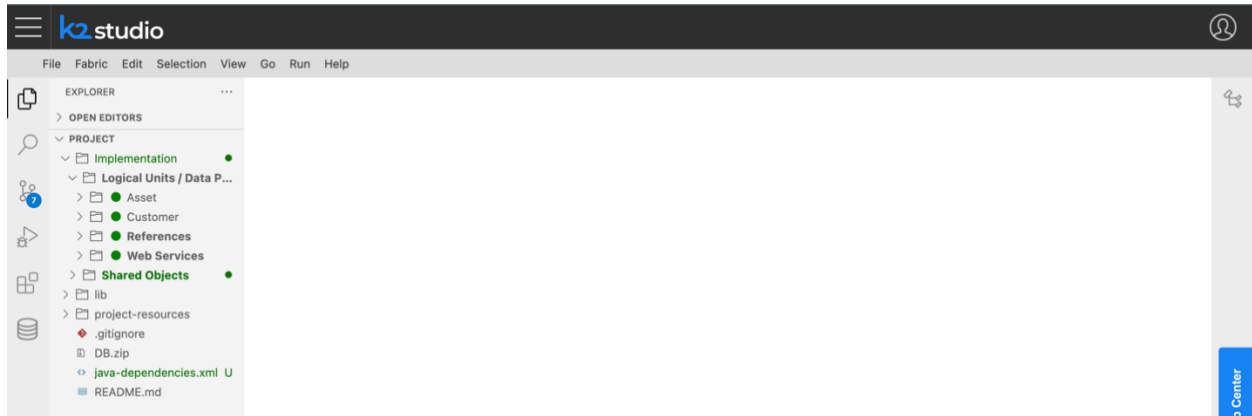
Note: Some organizations may not allow the use of DNS records created in your installation's Public DNS zone to propagate your organization's DNS. If that's the case, you must provide the address information to your organization. Refer to the [DNS zone](#) section for this.

7.1. Enter The Fabric Space

To enter your space, select the top left corner to launch the workspace.



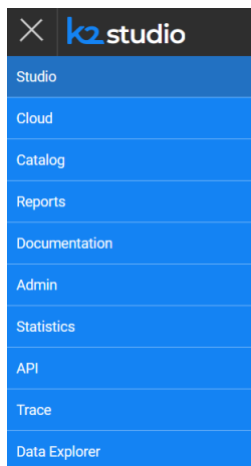
The **Studio** web application will be displayed.



Then, navigate to the various applications' context menus and confirm that each web application appears functional by selecting the “hamburger” menu item on the top left of the page.



The following is the list of web applications you should expect to see.



8. Completion

This completes the end of your installation and configuration of the K2view Azure Application, the creation of your K2view Project and the creation of your first K2view Project.

We invite you to visit our Knowledge Base at <https://support.k2view.com/knowledge-base.html>, where you can find documentation and demo projects, and sign up for the K2view Academy Learning Center at <https://academy.k2view.com>. For access, please talk to your K2view representative.

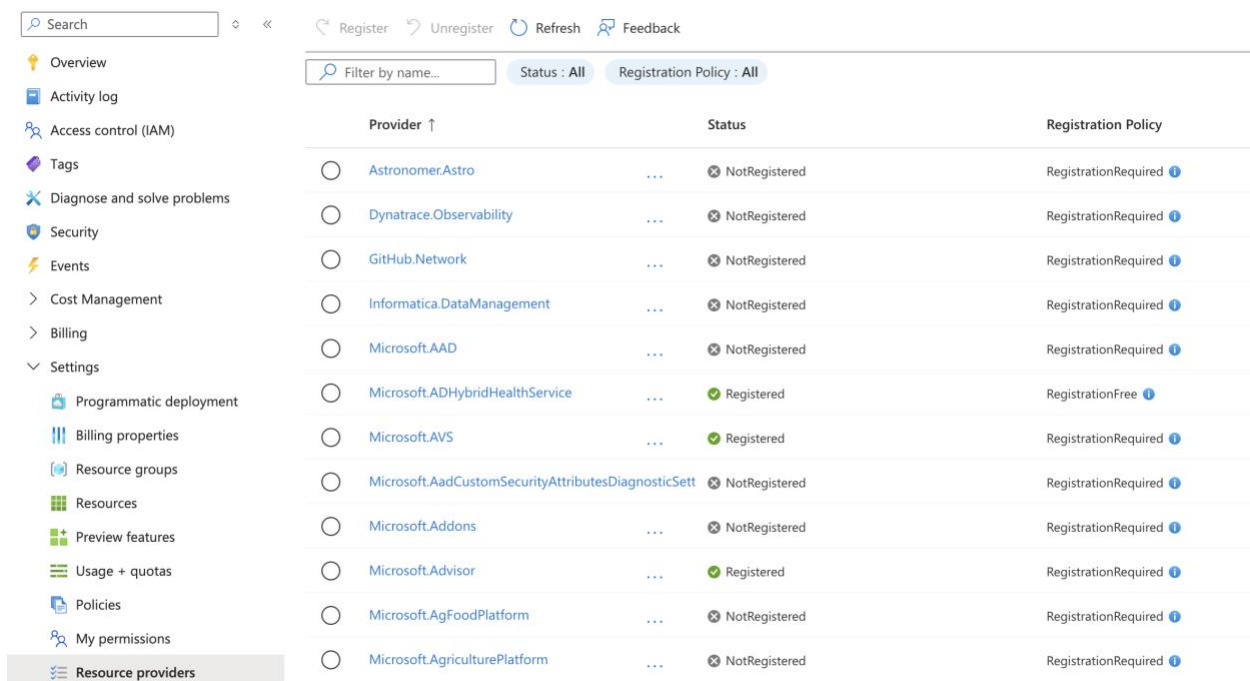
9. Instructions

9.1. Azure Resource Registration

The List of Azure Resources

An Azure resource provider is a set of REST operations that enable functionality for a specific Azure service. When deploying K2view Fabric services within an Azure environment, resource providers must be registered before the start of the installation.

The act of registering resources is a privileged action that needs to be performed by a Subscription Owner. This can be performed using the Azure UI at the subscription level as shown here, or automated using the Azure CLI (described in the next section).



Provider ↑	Status	Registration Policy
Astronomer.Astro	NotRegistered	RegistrationRequired
Dynatrace.Observability	NotRegistered	RegistrationRequired
GitHub.Network	NotRegistered	RegistrationRequired
Informatica.DataManagement	NotRegistered	RegistrationRequired
Microsoft.AAD	NotRegistered	RegistrationRequired
Microsoft.ADHybridHealthService	Registered	RegistrationFree
Microsoft.AVS	Registered	RegistrationRequired
Microsoft.AadCustomSecurityAttributesDiagnosticSett	NotRegistered	RegistrationRequired
Microsoft.Addons	NotRegistered	RegistrationRequired
Microsoft.Advisor	Registered	RegistrationRequired
Microsoft.AgFoodPlatform	NotRegistered	RegistrationRequired
Microsoft.AgriculturePlatform	NotRegistered	RegistrationRequired

The following list of Azure Resources required by K2view Fabric services need to be registered at the Subscription level.

- ManagedIdentity
- ContainerService
- Storage
- ContainerRegistry
- Maintenance
- CognitiveServices
- HDInsight
- Media
- ServiceBus
- SignalRService

- Search
- AppConfigurations
- SecurityInsights
- Maps
- DataLakeAnalytics
- Kusto
- Databricks
- StreamAnalytics
- DocumentDB
- MachineLearningServices
- EventGrid
- insights
- DBforPostgreSQL
- DBforMySQL
- OperationsManagement
- OperationalInsights
- Logic
- Compute
- ServiceFabric
- AppPlatform
- DesktopVirtualization
- PowerBIDedicated
- Cache
- GuestConfiguration
- BotService
- AVS
- DevTestLab
- DataMigration
- Cdn
- Web
- EventHub
- Security
- Network
- Automation
- Devices
- DBforMariaDB
- PolicyInsights
- ManagedServices
- Relay
- DataProtection
- CustomProviders

- Sql
- HealthcareApis
- Management
- DataFactory
- RecoveryServices
- MixedReality
- ApiManagement
- KeyVault
- NotificationHubs
- DataLakeStore

Automating Resource Registration Using the Azure CLI

The Azure CLI automates the above. Once installed and login using the “az login” command, you can paste the following within the Azure CLI window.

After a few minutes all required resources will have been registered. You can use the Azure UI (shown above) to track progress and a CLI command (az provider list --query="[?registrationState=='Registered']" --output table) to do so.

```
az provider register --namespace Microsoft.ManagedIdentity
az provider register --namespace Microsoft.ContainerService
az provider register --namespace Microsoft.Storage
az provider register --namespace Microsoft.ContainerRegistry
az provider register --namespace Microsoft.Maintenance
az provider register --namespace Microsoft.CognitiveServices
az provider register --namespace Microsoft.HDInsight
az provider register --namespace Microsoft.Media
az provider register --namespace Microsoft.ServiceBus
az provider register --namespace Microsoft.SignalRService
az provider register --namespace Microsoft.Search
az provider register --namespace Microsoft.AppConfiguration
az provider register --namespace Microsoft.SecurityInsights
az provider register --namespace Microsoft.Maps
az provider register --namespace Microsoft.DataLakeAnalytics
az provider register --namespace Microsoft.Kusto
az provider register --namespace Microsoft.Databricks
az provider register --namespace Microsoft.StreamAnalytics
az provider register --namespace Microsoft.DocumentDB
az provider register --namespace Microsoft.MachineLearningServices
az provider register --namespace Microsoft.EventGrid
az provider register --namespace microsoft.insights
az provider register --namespace Microsoft.DBforPostgreSQL
az provider register --namespace Microsoft.DBforMySQL
az provider register --namespace Microsoft.OperationsManagement
az provider register --namespace Microsoft.OperationInsights
az provider register --namespace Microsoft.Logic
az provider register --namespace Microsoft.Compute
az provider register --namespace Microsoft.ServiceFabric
az provider register --namespace Microsoft.AppPlatform
```

```
az provider register --namespace Microsoft.DesktopVirtualization
az provider register --namespace Microsoft.PowerBIDedicated
az provider register --namespace Microsoft.Cache
az provider register --namespace Microsoft.GuestConfiguration
az provider register --namespace Microsoft.BotService
az provider register --namespace Microsoft.AVS
az provider register --namespace Microsoft.DevTestLab
az provider register --namespace Microsoft.DataMigration
az provider register --namespace Microsoft.Cdn
az provider register --namespace Microsoft.Web
az provider register --namespace Microsoft.EventHub
az provider register --namespace Microsoft.Security
az provider register --namespace Microsoft.Network
az provider register --namespace Microsoft.Automation
az provider register --namespace Microsoft.Devices
az provider register --namespace Microsoft.DBforMariaDB
az provider register --namespace Microsoft.PolicyInsights
az provider register --namespace Microsoft.ManagedServices
az provider register --namespace Microsoft.Relay
az provider register --namespace Microsoft.DataProtection
az provider register --namespace Microsoft.CustomProviders
az provider register --namespace Microsoft.Sql
az provider register --namespace Microsoft.HealthcareApis
az provider register --namespace Microsoft.Management
az provider register --namespace Microsoft.DataFactory
az provider register --namespace Microsoft.RecoveryServices
az provider register --namespace Microsoft.MixedReality
az provider register --namespace Microsoft.ApiManagement
az provider register --namespace Microsoft.KeyVault
az provider register --namespace Microsoft.NotificationHubs
az provider register --namespace Microsoft.DataLakeStore
```

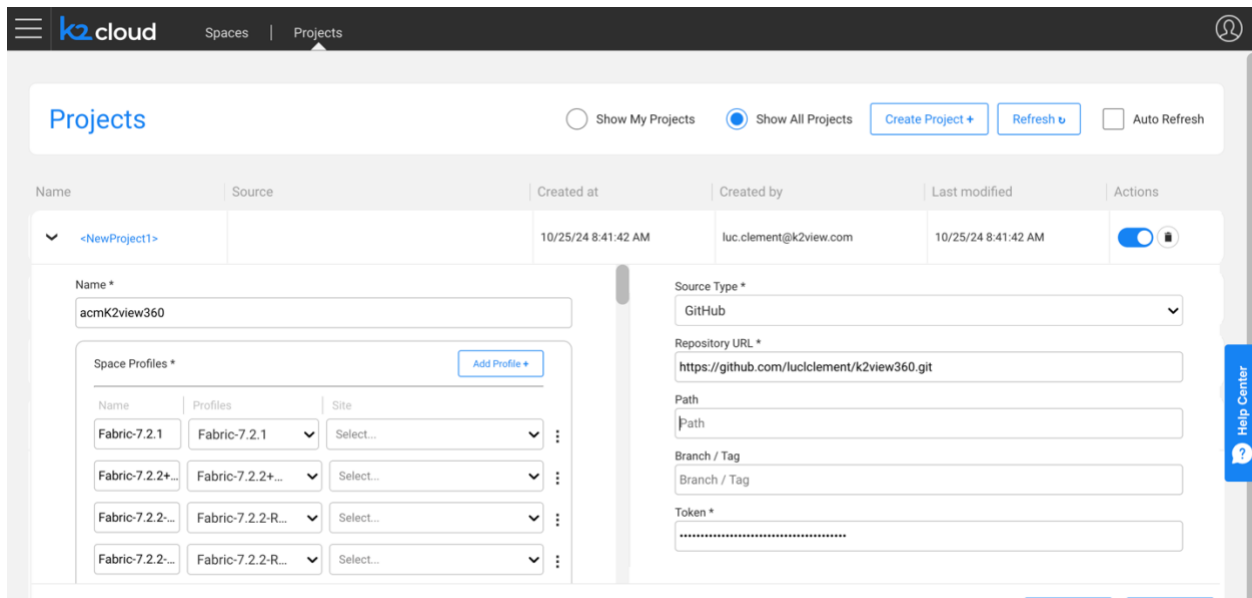
9.2. GitHub

Your organization may already have a Git repository. GitHub, GitLab and BitBucket are cloud providers that you can turn to should you like to get started.

During the course of this installation you will be asked to provide Git source code repository information including:

1. The provider: GitHub, GitLab, or BitBucket
2. A repository URL: in our example we used GitHub and this repository URL:
<https://github.com/luccllement/k2view360.git>
3. A branch: to keep things simple we used “main”
4. A Git Token: to provide the project access to the Git account

This information is captured on the project’s page



The screenshot shows the K2view cloud interface with the 'Projects' tab selected. The page displays a table of projects and a detailed configuration form for a new project named 'acmK2view360'.

Name	Source	Created at	Created by	Last modified	Actions
<NewProject1>		10/25/24 8:41:42 AM	luc.clement@k2view.com	10/25/24 8:41:42 AM	☒ ☐

Project Configuration Form:

- Name ***: acmK2view360
- Source Type ***: GitHub
- Repository URL ***: <https://github.com/luccllement/k2view360.git>
- Path**: Path
- Branch / Tag**: Branch / Tag
- Token ***: [Redacted]

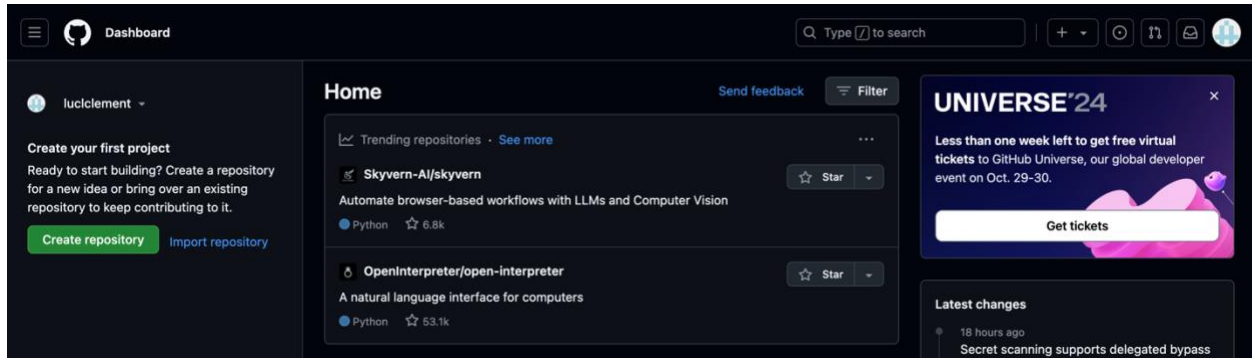
Space Profiles *

Name	Profiles	Site
Fabric-7.2.1	Fabric-7.2.1	Select...
Fabric-7.2.2+...	Fabric-7.2.2+...	Select...
Fabric-7.2.2-...	Fabric-7.2.2-R...	Select...
Fabric-7.2.2-...	Fabric-7.2.2-R...	Select...

If you don't have one you can obtain from github.com. First you need to sign up at <https://github.com>
You will create a repository and then obtain a token.

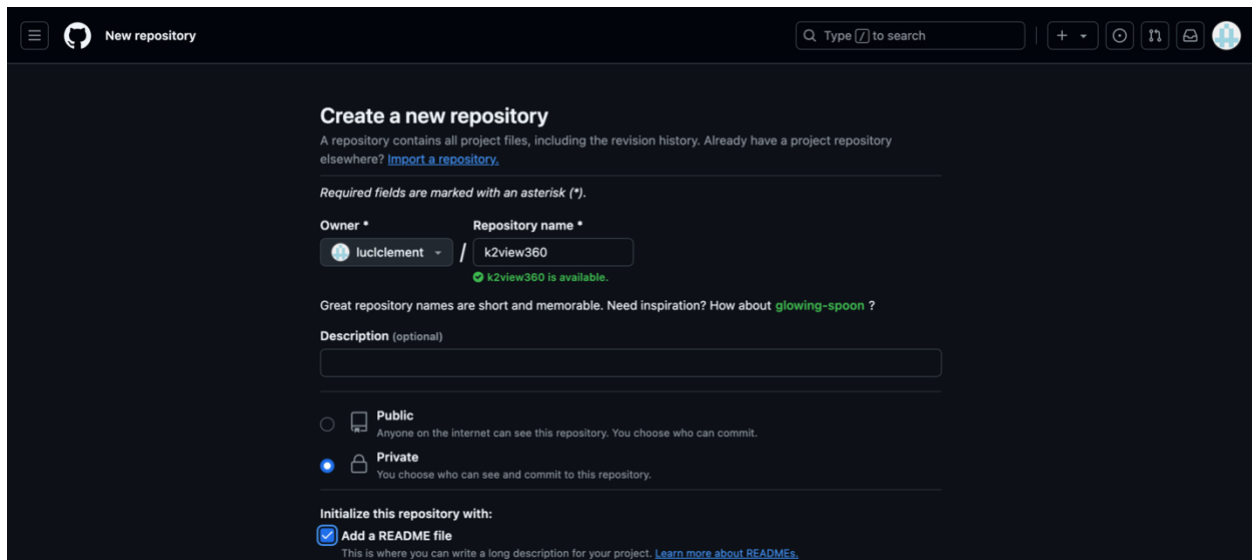
Create your repository

Click the “Create repository” button on the left side panel.

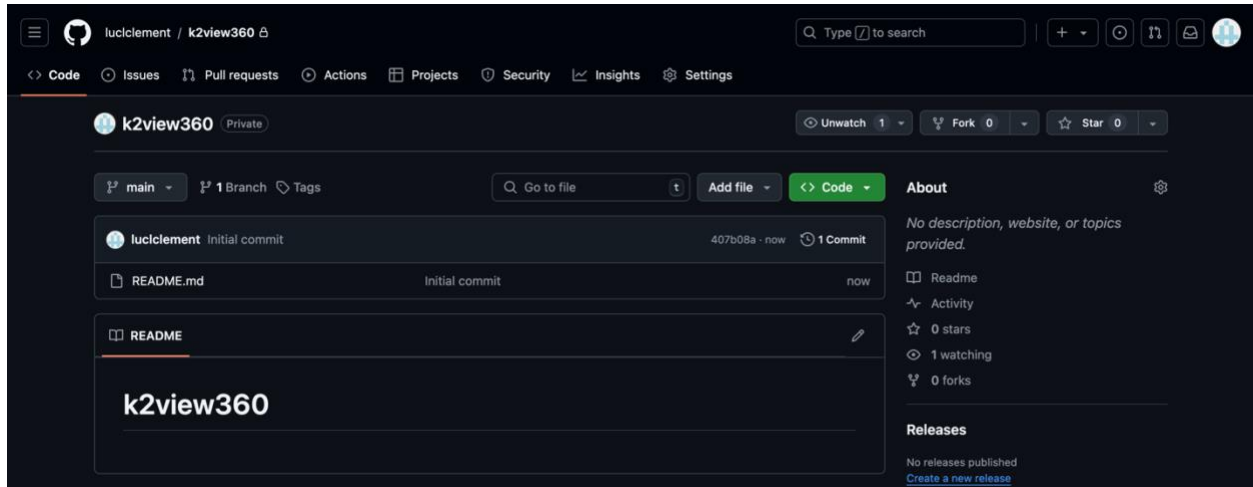


Give it a name. In this example we used k2view360.

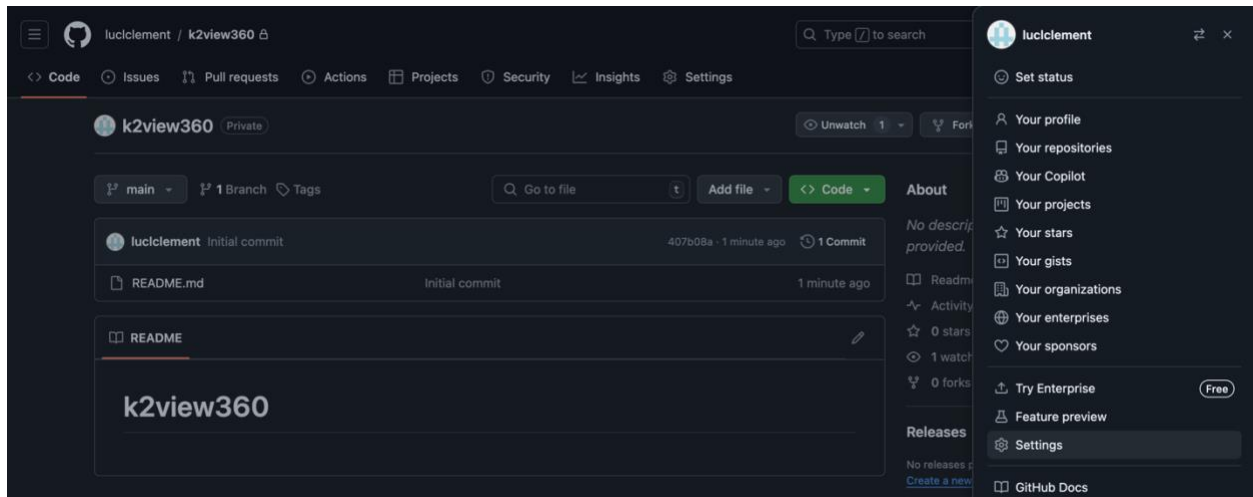
You should make it “Private”, and you should select the “Add a README file checkbox so that the repository is not empty.



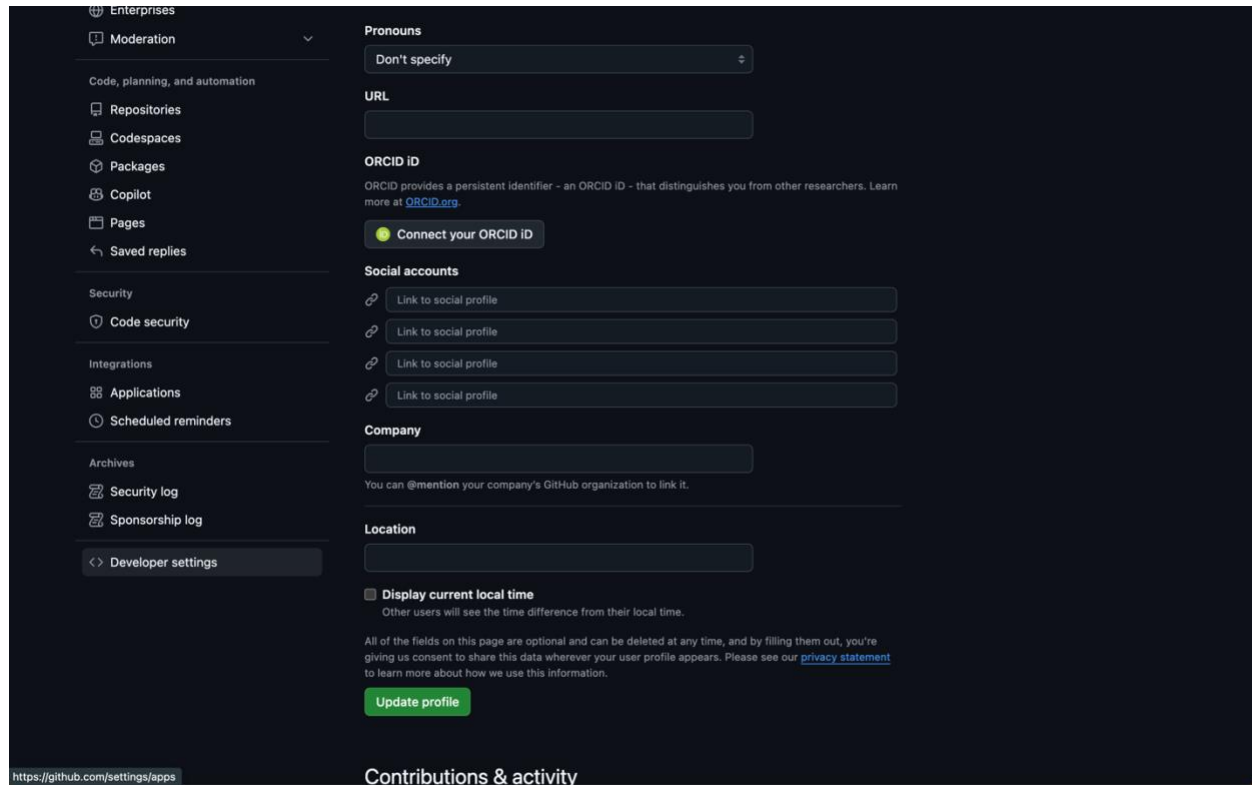
Once created you will find an empty README file.



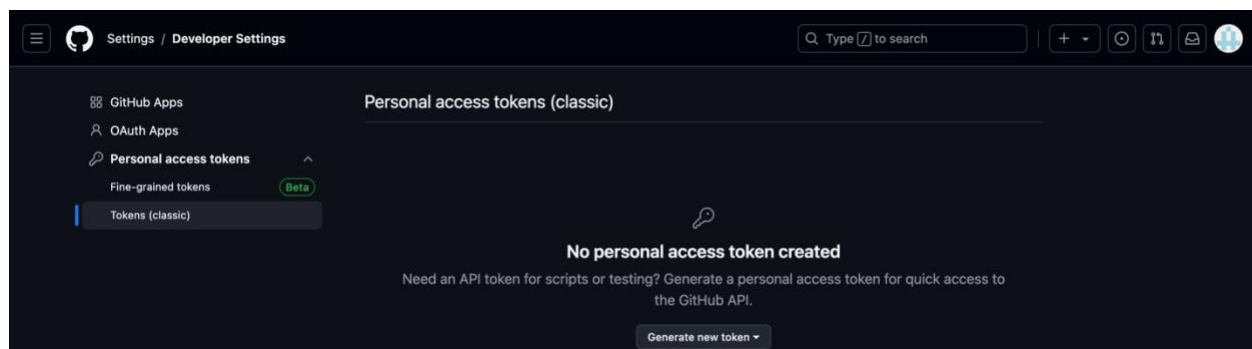
The next task is to obtain your Token. For this, drop down the user's icon on the top right of the page. Select Settings.



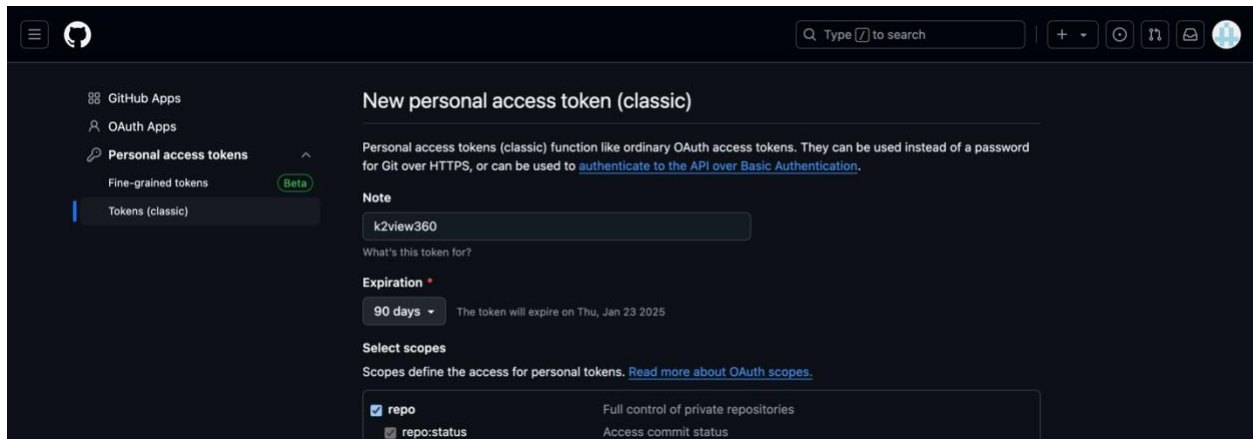
Then, from the left panel, select “Developer settings” on the bottom of the list.



Then, select “Personal access tokens” from the left panel, and select “Tokens (classic)” (if you’re adventurous, try the beta).



Select the “Generate new token” on the bottom of the page and create your token.



When the page refreshes you will be shown your token. Copy it to use in the Fabric Project creation page. Keep a copy of the token for safekeeping.

9.3. DNS zone

During the installation, a DNS zone will be created. This zone may take time to propagate into your environment and may not be available until it does.

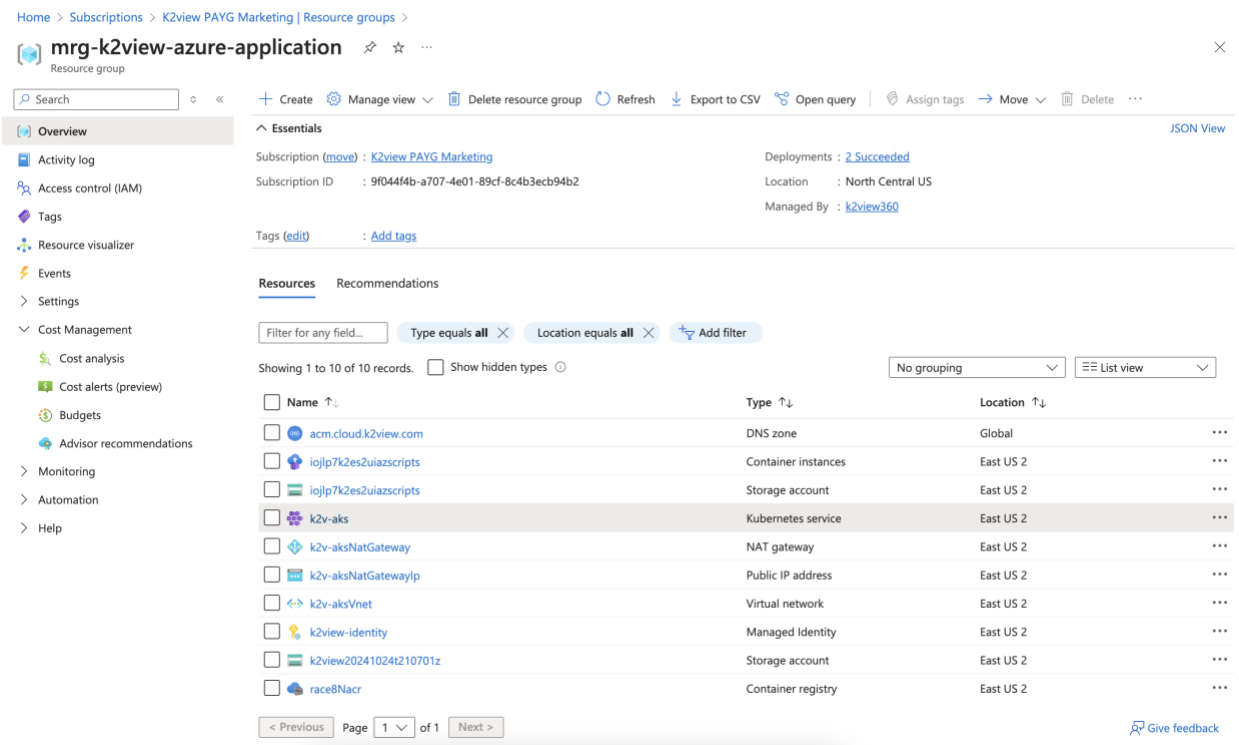
Some organizations may not allow the use of DNS record created in your installation's Public DNS zone to propagate outside the organization.

If that's the case, you will need to provide the address information to your organization to generate a DNS A record for the NGINX Ingress controller load balancer IP address created in our Public DNS Zone.

You should note that when creating a namespace, its name is associated as a subdomain to this domain name in the Ingress controller. For example, if the domain is "k2dev.company.com" and a created namespace is "test", then the URL of this namespace that users shall access will be "test.k2dev.company.com".

In our example, we will connect to a space accessible at acmk2view360-k2v-rnd.acm.cloud.k2view.com, a publicly resolvable domain.

To locate the address, select the Kubernetes service



Home > Subscriptions > K2view PAYG Marketing | Resource groups >

mrg-k2view-azure-application Resource group

Search

+ Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Move Delete

Overview Essentials JSON View

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Cost analysis

Cost alerts (preview)

Budgets

Advisor recommendations

Monitoring

Automation

Help

Subscription (move): [K2view PAYG Marketing](#) Deployments: [2 Succeeded](#)

Subscription ID: 9f044f4b-a707-4e01-89cf-8c4b3ecb94b2 Location: North Central US

Managed By: [k2view360](#)

Tags (edit): Add tags

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Showing 1 to 10 of 10 records. Show hidden types

No grouping List view

Name	Type	Location
acm.cloud.k2view.com	DNS zone	Global
iojlp7k2es2uiaazscripts	Container instances	East US 2
iojlp7k2es2uiaazscripts	Storage account	East US 2
k2v-aks	Kubernetes service	East US 2
k2v-aksNatGateway	NAT gateway	East US 2
k2v-aksNatGatewayIp	Public IP address	East US 2
k2v-aksVnet	Virtual network	East US 2
k2view-identity	Managed Identity	East US 2
k2view202410241210701z	Storage account	East US 2
race8Nacr	Container registry	East US 2

< Previous Page 1 of 1 Next >

Give feedback

Select the Services and ingresses. The address you are looking for is the ingress-nginx-controller's load balancer. In this example the address is 48.214.88.212.

Home > Subscriptions > K2view PAYG Marketing | Resource groups > mrg-k2view-azure-application > k2v-aks

k2v-aks | Services and ingresses ☆ ...

Kubernetes service

Search [] Create Delete Refresh Show labels Give feedback

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Microsoft Defender for Cloud
Cost analysis
Kubernetes resources
Namespaces
Workloads
Services and ingresses ☆
Storage
Configuration
Custom resources
Events
Run command
Settings
Monitoring
Automation
Help

Services Ingresses

Filter by service name: Enter the full service name
Filter by namespace: All namespaces Add label filter

☐	Name	Namespace	Status	Type	Cluster IP	External IP	Ports	Age	
☐	kubernetes	default	Ok	ClusterIP	10.0.0.1		443/TCP	2 days	...
☐	kube-dns	kube-system	Ok	ClusterIP	10.0.0.10		53/UDP53/TCP	2 days	...
☐	metrics-server	kube-system	Ok	ClusterIP	10.0.66.7		443/TCP	2 days	...
☐	ingress-nginx-controller	ingress-nginx	Ok	LoadBalancer	10.0.149.85	48.214.88.212	80:31263/TCP4...	2 days	...
☐	ingress-nginx-controller...	ingress-nginx	Ok	ClusterIP	10.0.132.98		443/TCP	2 days	...
☐	ingress-test-service	ingress-nginx	Ok	ClusterIP	10.0.68.142		5678/TCP	2 days	...
☐	nginx-errors	ingress-nginx	Ok	ClusterIP	10.0.10.253		80/TCP	2 days	...
☐	fabric-service	acmk2view360-k2v-md	Ok	ClusterIP	10.0.55.51		3213/TCP	1 day	...

https://portal.azure.com/#@k2viewrmd.onmicrosoft.com/resource/subscriptions/9f04df4b-a707-4e01-88cf-8c4b3e6b9d52/resourceGroups/mrg-k2view-azure-application/providers/Microsoft.ContainerService/managedClusters/k2v-aks/services/

9.4. Certificates

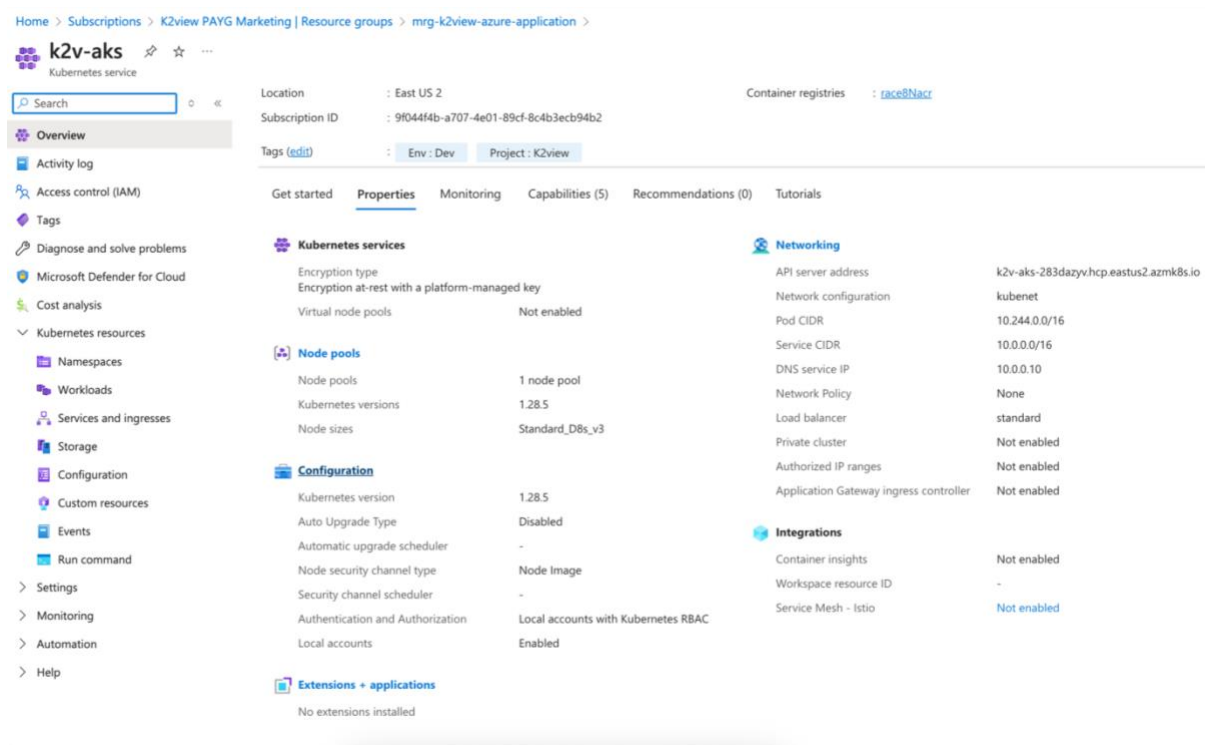
Expiration

NOTE: Certificates have expiration dates. Please note the date of expiration of the certificate provided to you and replace it before the expiration date as described next.

Replacing Certificates

If you encounter certificate issues or need to update your certificate, follow these instructions. You can also make a support request with K2view for assistance.

First, select the Managed Azure Application resource group created by the installation. Select the Kubernetes service item.



The screenshot shows the Azure portal interface for a Kubernetes service named 'k2v-aks'. The breadcrumb navigation indicates the path: Home > Subscriptions > K2view PAYG Marketing | Resource groups > mrg-k2view-azure-application > k2v-aks. The left sidebar contains a search bar and a list of navigation options including Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Microsoft Defender for Cloud, Cost analysis, and various Kubernetes resources like Namespaces, Workloads, Services and ingresses, Storage, Configuration, Custom resources, Events, Run command, Settings, Monitoring, Automation, and Help.

The main content area displays the 'Properties' tab for the 'k2v-aks' service. It shows the following details:

- Location:** East US 2
- Subscription ID:** 9f044f4b-a707-4e01-89cf-8c4b3ecb94b2
- Container registries:** [raced@Nacr](#)
- Tags:** Env: Dev, Project: K2view

Below these details, there are several sections for configuration and monitoring:

- Get started:** Properties (selected), Monitoring, Capabilities (5), Recommendations (0), Tutorials.
- Kubernetes services:**
 - Encryption type: Encryption at-rest with a platform-managed key
 - Virtual node pools: Not enabled
- Node pools:**
 - Node pools: 1 node pool
 - Kubernetes versions: 1.28.5
 - Node sizes: Standard_D8s_v3
- Configuration:**
 - Kubernetes version: 1.28.5
 - Auto Upgrade Type: Disabled
 - Automatic upgrade scheduler: -
 - Node security channel type: Node Image
 - Security channel scheduler: -
 - Authentication and Authorization: Local accounts with Kubernetes RBAC
 - Local accounts: Enabled
- Extensions + applications:** No extensions installed
- Networking:**
 - API server address: k2v-aks-283dazyvhcp.eastus2.azmk8s.io
 - Network configuration: kubenet
 - Pod CIDR: 10.244.0.0/16
 - Service CIDR: 10.0.0.0/16
 - DNS service IP: 10.0.0.10
 - Network Policy: None
 - Load balancer: standard
 - Private cluster: Not enabled
 - Authorized IP ranges: Not enabled
 - Application Gateway ingress controller: Not enabled
- Integrations:**
 - Container insights: Not enabled
 - Workspace resource ID: -
 - Service Mesh - Istio: Not enabled

Then select the “Configuration” left menu item as shown here, select the Secrets tab, and then select the “wildcard-certificate”.

Home > Subscriptions > k2view PAYG Marketing | Resource groups > mrg-k2view-azure-application > k2v-aks

k2v-aks | Configuration ☆ ...

Kubernetes service

Search Create Delete Refresh Show labels Give feedback

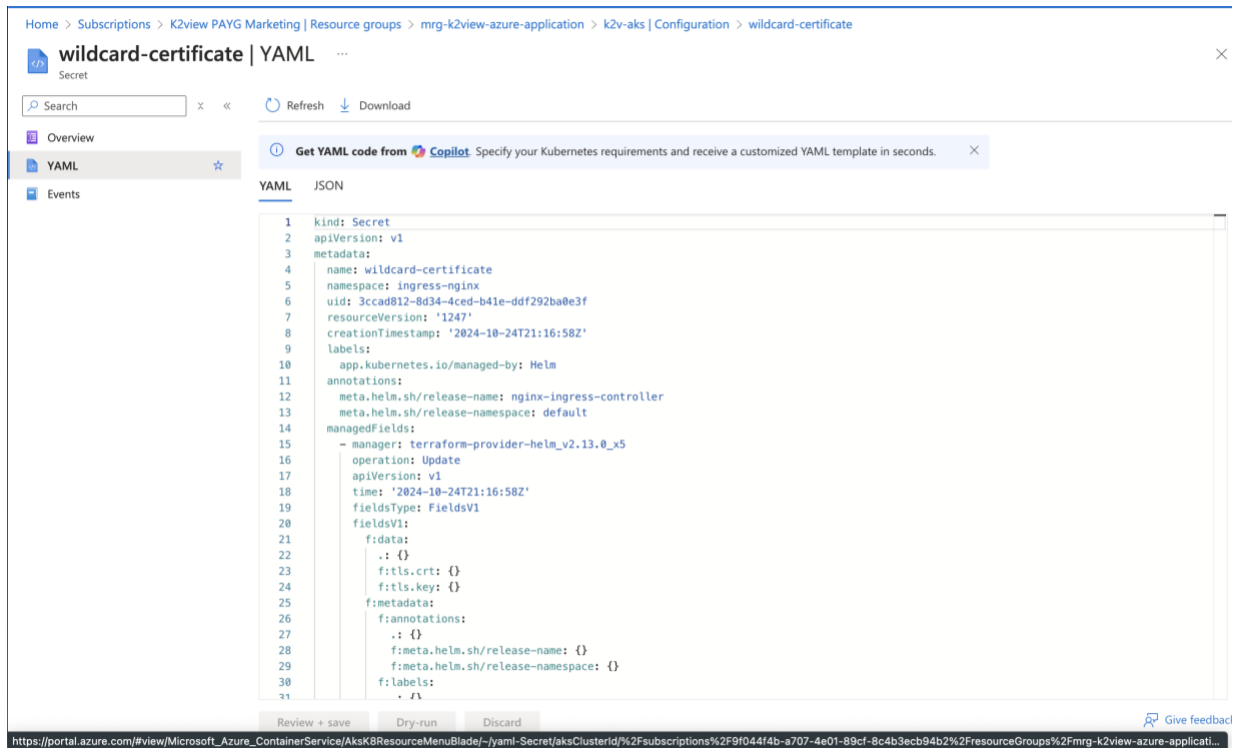
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Microsoft Defender for Cloud
Cost analysis
Kubernetes resources
Namespaces
Workloads
Services and ingresses
Storage
Configuration
Custom resources
Events
Run command
Settings
Node pools
Cluster configuration
Application scaling
Networking
Extensions + applications

Config maps **Secrets**

Filter by secret name Filter by namespace Add label filter

☐	Name	Namespace	Type	Data	Age
☐	connectivity-certs	kube-system	Opaque	3	22 hours
☐	bootstrap-token-76dr0f	kube-system	bootstrap.kubernetes.io...	4	22 hours
☐	sh.helm.release.v1.k2view-agent.v1	default	helm.sh/release.v1	1	22 hours
☐	agent-config-secrets	k2view-agent	Opaque	5	22 hours
☐	k2view-agent-secret	k2view-agent	kubernetes.io/service-ac...	3	22 hours
☐	sh.helm.release.v1.nginx-ingress-controller.v1	default	helm.sh/release.v1	1	22 hours
☐	wildcard-certificate	ingress-nginx	kubernetes.io/tls	2	22 hours
☐	ingress-nginx-admission	ingress-nginx	Opaque	3	22 hours
☐	common-env-secrets	acmk2view360-k2v-rnd	Opaque	12	3 hours
☐	config-init-secrets	acmk2view360-k2v-rnd	Opaque	6	3 hours
☐	config-secrets	acmk2view360-k2v-rnd	Opaque	5	3 hours
☐	secret-deploy-apikey	acmk2view360-k2v-rnd	Opaque	1	3 hours

After opening the wildcard certificate and selecting the YAML section, scroll down and locate the Key and Certificate sections and paste replacement values (not shown here). These values are the base64 encoded values of your PEM certificate and private key.



Your PEM TLS certificate and its associated private key

Your network team must create a wildcard TLS certificate for the domain you selected as your Domain Ingress. The full certificate authority chain needs to be present. During the installation, you must paste the base64-encoded values of your TLS certificate and its associated private key.

- **TLS Private:** the base64-encoded private key associated with the TLS certificate
- **TLS Certificate:** a base64-encoded wildcard certificate. For example, for the domain ingress of acm.cloud.k2view.com you need to create a wildcard certificate for *.acm.cloud.k2view.com for hosts of this domain.

What you paste in should look like the following:

base64-encoded TLS Private Key:

```
LS0tLS1CRUdJTiBQUk1wQVRFIEtFWS0tLS0tCk1JSUpRd01CQURBTk1na3Foa21HOXcwQkFRRUZBQVNDQ1Mwd2dna3BBZ  
OVBQW9JJQ0FRRE9yakUrNONQVku2QVgKV01FRFVvdnVlc01XOEJ4MTR  
L...  
QT1ZV1pKKz1qRkhFUTdBZXB3S1RtMEVraz0KLS0tLS1FTkQgUjJkFURSBLRVktLS0tLQo=
```

base64-encoded TLS Certificate:

```
LS0tLS1CRUdJTiBDRVJUSUZJQ0FURSB0tLS0tCk1JSUdGRENDQ1B5Z0F3SUJBZ01TQS9YVFJkaUpXOXMzb1dZZ1N5a0RpZ  
1h5TUeWR0NTcUdTSWlZRFFFQkN3VUEKTURNeEN6QUpcZ05WQkFZVEFsV1RNU113RkFZRFZRUUtFdzFNW1hRbm  
...  
1ND0KLS0tLS1FTkQgQ0VSVE1GSUNBVEUtLS0tLQ==
```

Note: The PEM certificate and key must be further encoded using base64 as input to the installation screen. To encode a PEM certificate (e.g., cert.pem) to base64 on both Linux and Windows, you can use the following commands.

Linux: You can use the base64 command to read the cert.pem file, encodes it in base64, and saves the output to cert_base64.txt.

```
base64 cert.pem > cert_base64.txt
```

Windows: On Windows, you can use the certutil command to encode the cert.pem file in base64 and saves the result to cert_base64.txt.

```
certutil -encode cert.pem cert_base64.txt
```

Prerequisite: You should prepare these base64 encoded files before the start of the installation.

10. Troubleshooting

DNS Issues

To start, please refer to the [DNS zone](#) section. For assistance, please make a support request with K2view.